Causal discovery for robust predictions

Homer Durand

Image Processing Lab (ISP group)

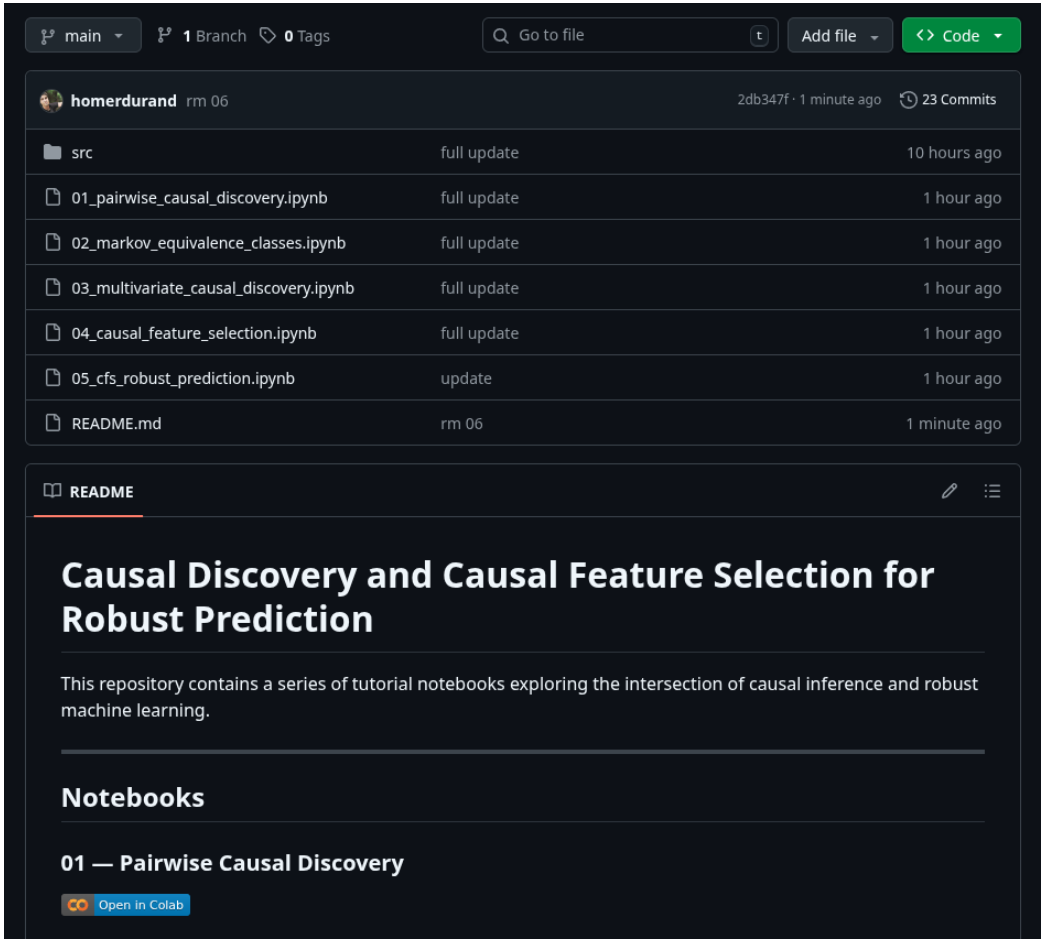
Mar 2026

0 Outline

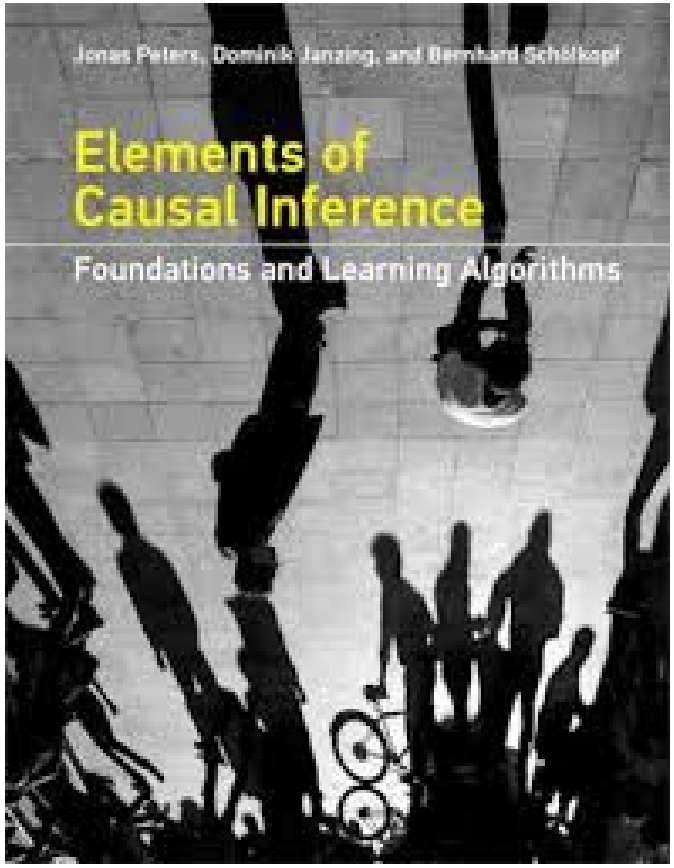
1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

0 Github repo

github.com/WinterSchool2026/zh03-causal-discovery-robust-predictions

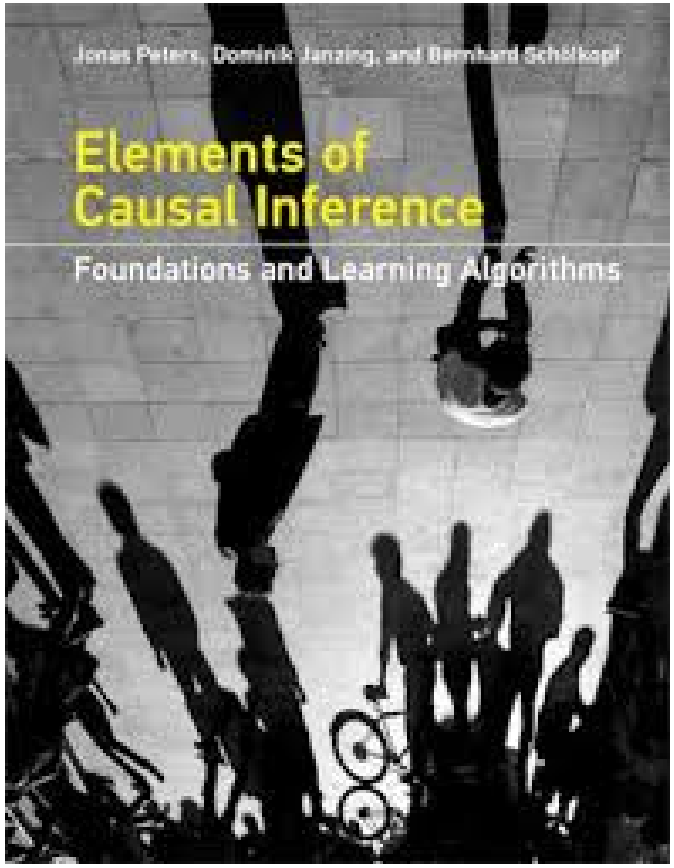


0 Literature - Statistical Causality

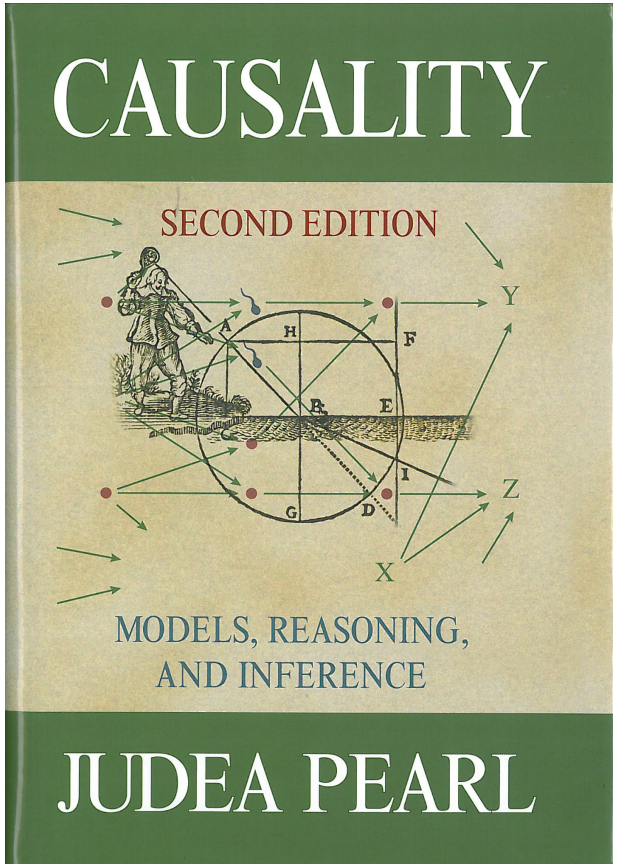


Peters, Janzing, & Schölkopf
Elements of Causal Inference (2017)

0 Literature - Statistical Causality



Peters, Janzing, & Schölkopf
Elements of Causal Inference (2017)



Judea Pearl
Causality (2009)

1 Outline

1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

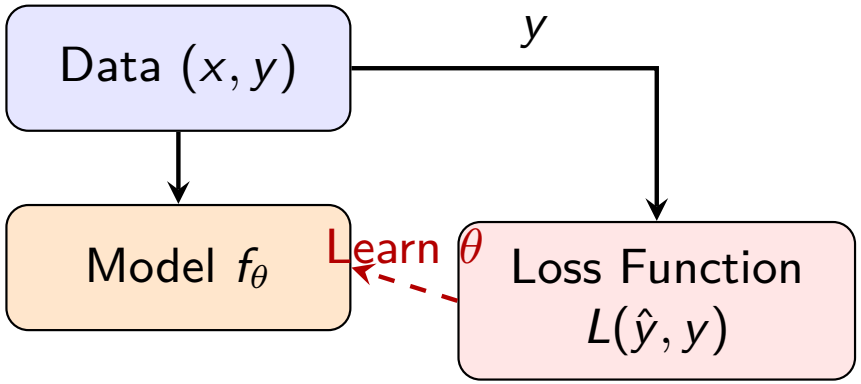
The basic recipe:

1. Collect **labelled data** $\{(x_i, y_i)\}_{i=1}^{n}$

Data (x, y)

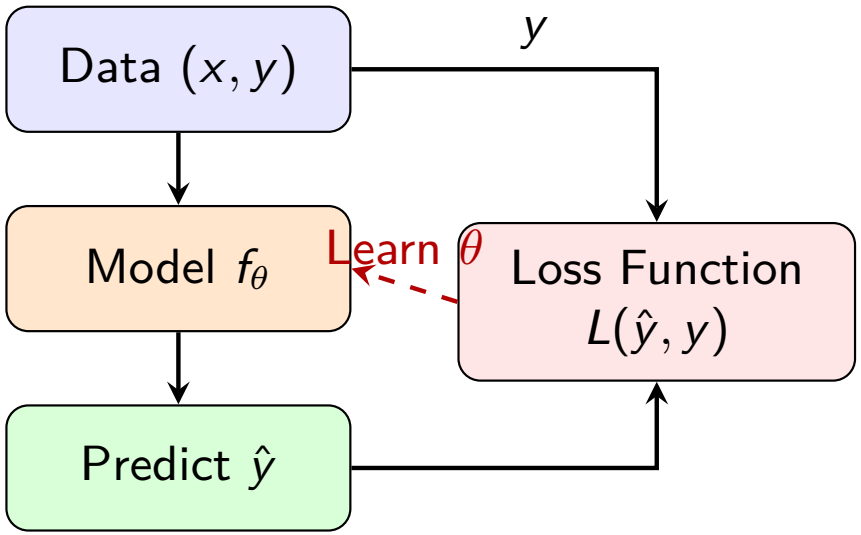
The basic recipe:

1. Collect **labelled data** $\{(x_i, y_i)\}_{i=1}^{n}$
2. **Train** a model f_θ that maps inputs x to outputs y



The basic recipe:

1. Collect **labelled data** $\{(x_i, y_i)\}_{i=1}^{n}$
2. **Train** a model f_θ that maps inputs x to outputs y
3. **Predict** on new data x_{n+1}



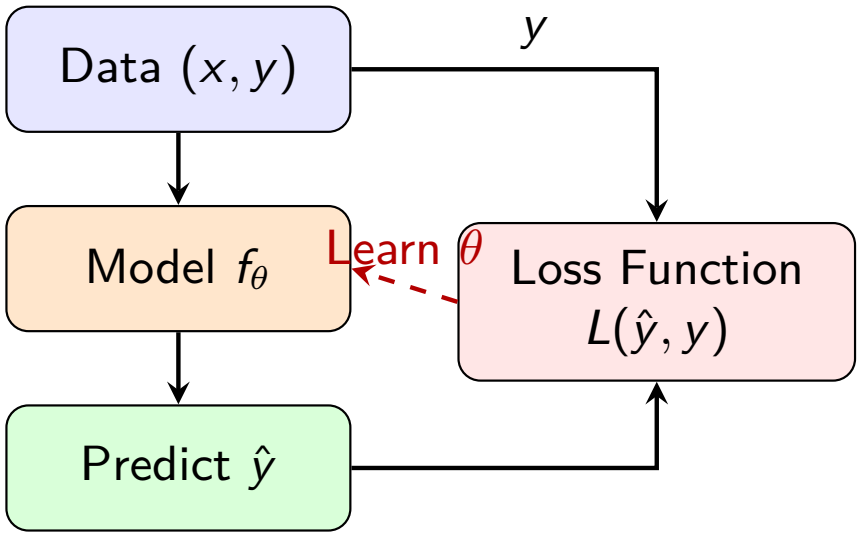
1 Machine Learning 101

The basic recipe:

1. Collect **labelled data** $\{(x_i, y_i)\}_{i=1}^{n}$
2. **Train** a model f_θ that maps inputs x to outputs y
3. **Predict** on new data x_{n+1}

Example:

- x = satellite observations (SST, humidity, ...)
- y = precipitations
- f_θ = Convolutional Neural Network
- L = Mean Squared Error



1 Empirical Risk Minimisation (ERM)

ERM: find parameters $\hat{\theta}$ minimising average prediction error over training data:

$$\hat{\theta} = \arg\min_{\theta} \underbrace{\mathbb{E}\big[L\big(y_i, f_\theta(x_i)\big)\big]}_{\text{training loss}}$$

1 Empirical Risk Minimisation (ERM)

ERM: find parameters $\hat{\theta}$ minimising average prediction error over training data:

$$\hat{\theta} = \arg\min_{\theta} \underbrace{\mathbb{E}\big[L\big(y_i, f_\theta(x_i)\big)\big]}_{\text{training loss}}$$

Why does this work well?

1 Empirical Risk Minimisation (ERM)

ERM: find parameters $\hat{\theta}$ minimising average prediction error over training data:

$$\hat{\theta} = \arg\min_{\theta} \underbrace{\mathbb{E}\big[L\big(y_i, f_\theta(x_i)\big)\big]}_{\text{training loss}}$$

Why does this work well?

Core assumption: i.i.d. data

Training *and* test data are drawn from the **same** distribution: $P_{\text{train}} = P_{\text{test}}$.

1 Empirical Risk Minimisation (ERM)

ERM: find parameters $\hat{\theta}$ minimising average prediction error over training data:

$$\hat{\theta} = \arg\min_{\theta} \underbrace{\mathbb{E}\big[L\big(y_i, f_\theta(x_i)\big)\big]}_{\text{training loss}}$$

Why does this work well?

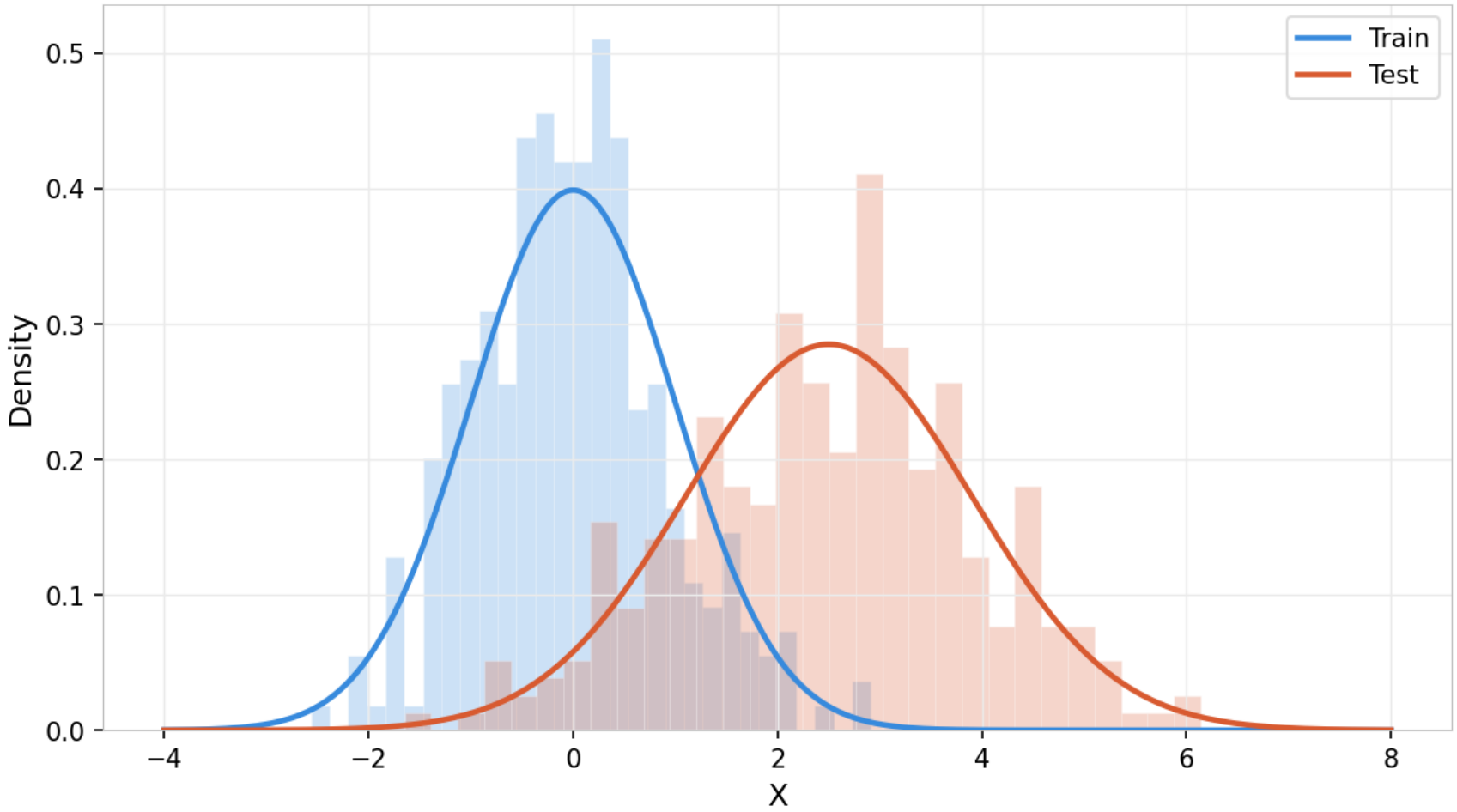
Core assumption: i.i.d. data

Training *and* test data are drawn from the **same** distribution: $P_{\text{train}} = P_{\text{test}}$.

Optimality of ERM

Assuming i.i.d data, ERM is optimal: it minimizes the expected risk.

1 Distribution shifts



Distribution shifts, $P_{\text{train}} \neq P_{\text{test}}$, are very common in climate science.

Distribution shifts, $P_{\text{train}} \neq P_{\text{test}}$, are very common in climate science.

Spatial shift
Model trained over one spatial regions
and tested in another one.

Distribution shifts, $P_{\text{train}} \neq P_{\text{test}}$, are very common in climate science.

Spatial shift
Model trained over one spatial regions and tested in another one.

Temporal shift
Trained on 1980–2000 data, tested on 2050 projections.
Long-term climate change make distribution shifts

Distribution shifts, $P_{\text{train}} \neq P_{\text{test}}$, are very common in climate science.

Spatial shift
Model trained over one spatial regions and tested in another one.

Temporal shift
Trained on 1980–2000 data, tested on 2050 projections.
Long-term climate change make distribution shifts

Forcing scenarios
RCP2.6 vs RCP8.5: CO_2 concentration is externally *set* — a true intervention on the climate system.

Distribution shifts, $P_{\text{train}} \neq P_{\text{test}}$, are very common in climate science.

Spatial shift
Model trained over one spatial regions and tested in another one.

Temporal shift
Trained on 1980–2000 data, tested on 2050 projections.
Long-term climate change make distribution shifts

Forcing scenarios
RCP2.6 vs RCP8.5: CO_2 concentration is externally *set* — a true intervention on the climate system.

Tipping points
AMOC slowdown, permafrost feedback: physical *mechanisms* themselves change

1 Robust ML

Idea: instead of minimising error on *one* distribution P, minimise error on the *worst case* over a family $\mathcal{P}$ of plausible distributions.

1 Robust ML

Idea: instead of minimising error on *one* distribution P, minimise error on the *worst case* over a family $\mathcal{P}$ of plausible distributions.

$$\theta^\star = \arg\min_\theta \underbrace{\sup_{P \in \mathcal{P}} \mathbb{E}_P\big[(Y - f_\theta(X))^2\big]}_{\text{worst-case loss}}$$

1 Robust ML

Idea: instead of minimising error on *one* distribution P, minimise error on the *worst case* over a family $\mathcal{P}$ of plausible distributions.

$$\theta^\star = \arg\min_\theta \underbrace{\sup_{P \in \mathcal{P}} \mathbb{E}_P\big[(Y - f_\theta(X))^2\big]}_{\text{worst-case loss}}$$

- $\mathcal{P}$ encodes **which shifts we expect** (e.g. different forcing scenarios).

1 Robust ML

Idea: instead of minimising error on *one* distribution P, minimise error on the *worst case* over a family $\mathcal{P}$ of plausible distributions.

$$\theta^\star = \arg\min_\theta \underbrace{\sup_{P \in \mathcal{P}} \mathbb{E}_P\big[(Y - f_\theta(X))^2\big]}_{\text{worst-case loss}}$$

- $\mathcal{P}$ encodes **which shifts we expect** (e.g. different forcing scenarios).
- Larger $\mathcal{P} \Rightarrow$ more robust, but *less accurate* in any single scenario.

1 Robust ML

Idea: instead of minimising error on *one* distribution P, minimise error on the *worst case* over a family $\mathcal{P}$ of plausible distributions.

$$\theta^\star = \arg\min_\theta \underbrace{\sup_{P \in \mathcal{P}} \mathbb{E}_P\big[(Y - f_\theta(X))^2\big]}_{\text{worst-case loss}}$$

- $\mathcal{P}$ encodes **which shifts we expect** (e.g. different forcing scenarios).
- Larger $\mathcal{P} \Rightarrow$ more robust, but *less accurate* in any single scenario.
- What makes some feature more robust?

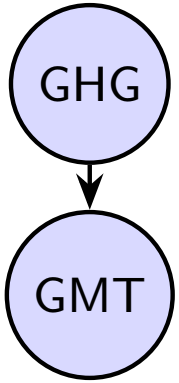
  $\Rightarrow$ **Causal structure!**

2 Outline

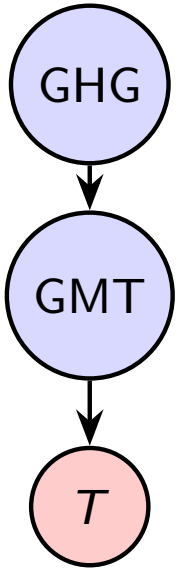
1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

A Directed Acyclic Graph (DAG):

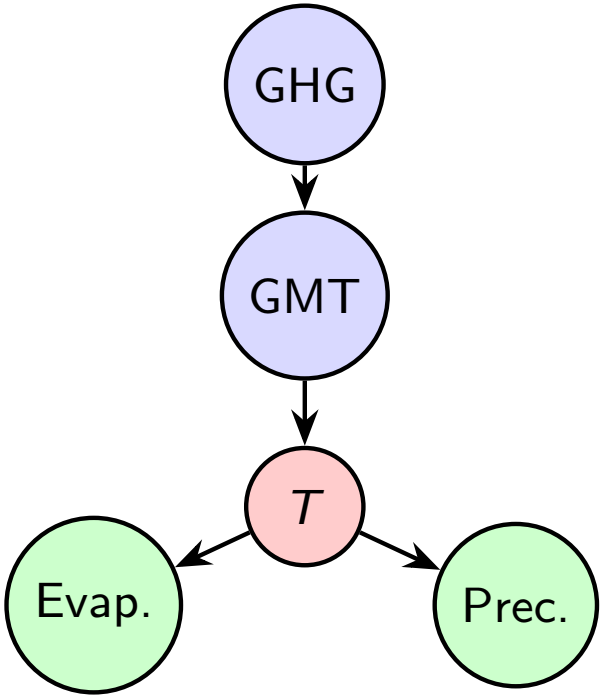
A Directed Acyclic Graph (DAG):



A Directed Acyclic Graph (DAG):

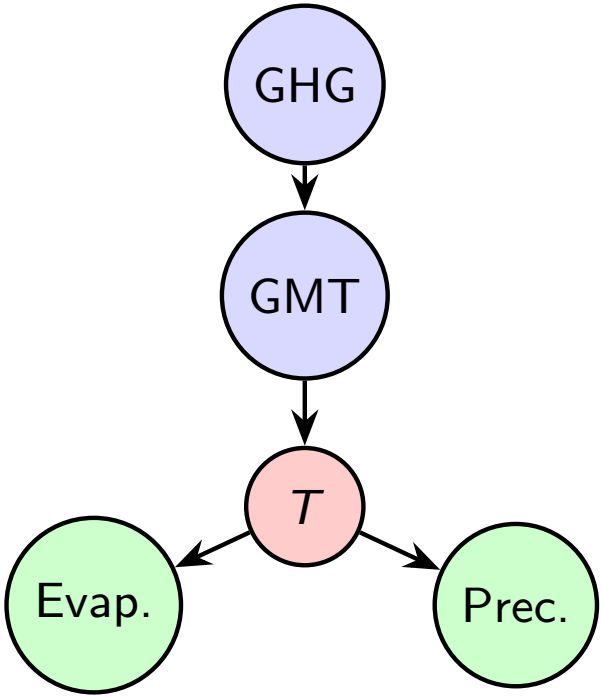


A Directed Acyclic Graph (DAG):



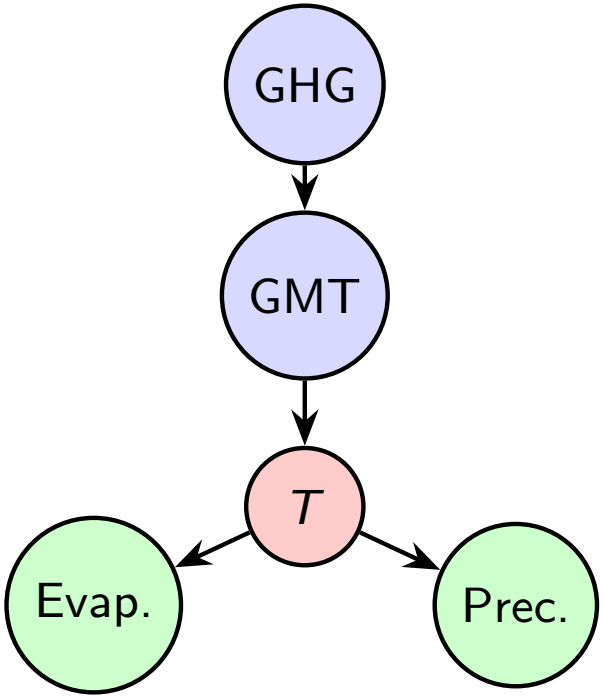
A Directed Acyclic Graph (DAG):

- Each **node** = a variable



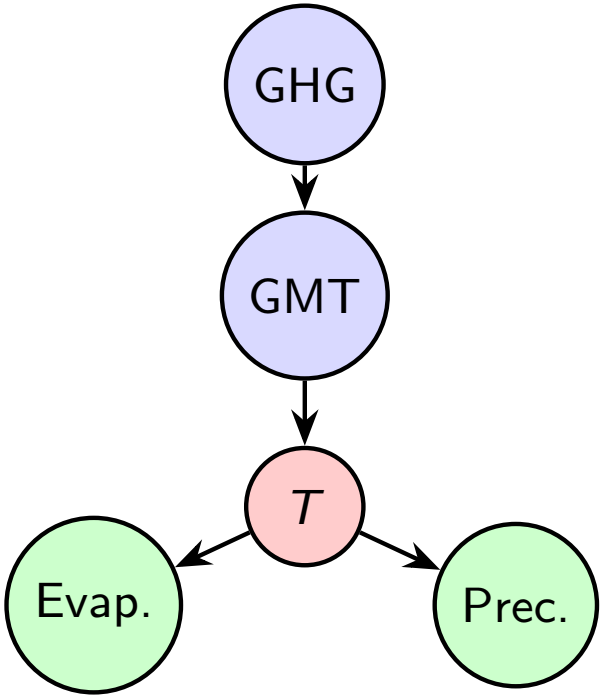
A Directed Acyclic Graph (DAG):

- Each **node** = a variable
- Each **arrow** $X \to Y$ = "X directly causes Y"



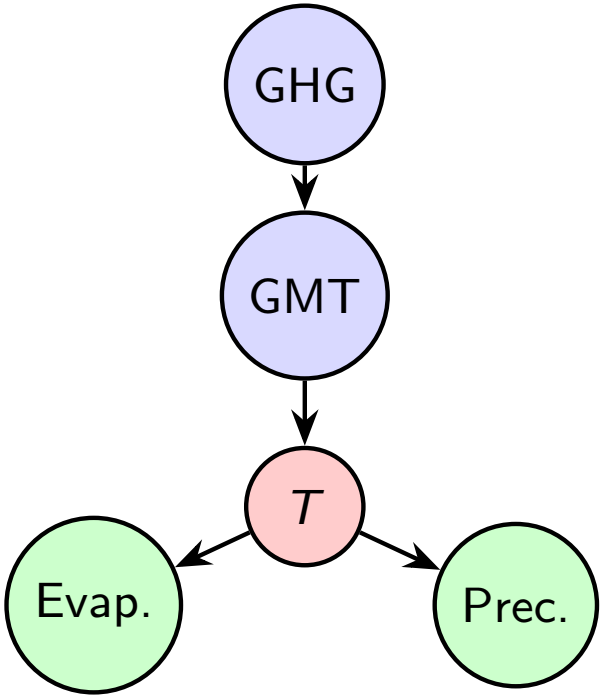
A Directed Acyclic Graph (DAG):

- Each **node** = a variable
- Each **arrow** $X \to Y$ = "X directly causes Y"
- The *mechanisms* between nodes are modular (ICM principle)



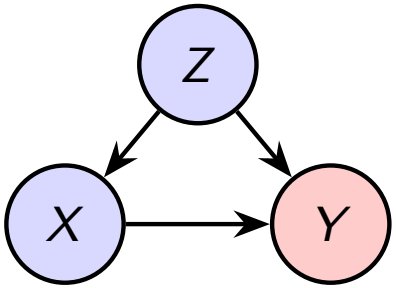
A Directed Acyclic Graph (DAG):

- Each **node** = a variable
- Each **arrow** $X \to Y$ = "X directly causes Y"
- The *mechanisms* between nodes are modular (ICM principle)
- No cycles



An intervention $do(X = x)$ sets a variable to a fixed value, overriding its natural causes, i.e. its **parents** in the graph.

Observational world

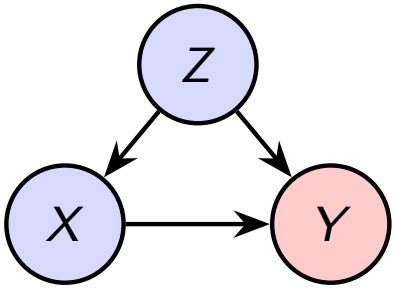


Z confounds X–Y

2 Interventions: Cutting Arrows in the Graph

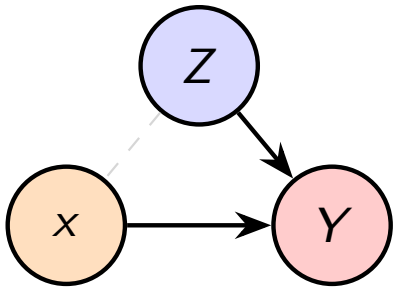
An intervention $do(X = x)$ sets a variable to a fixed value, overriding its natural causes, i.e. its **parents** in the graph.

Observational world



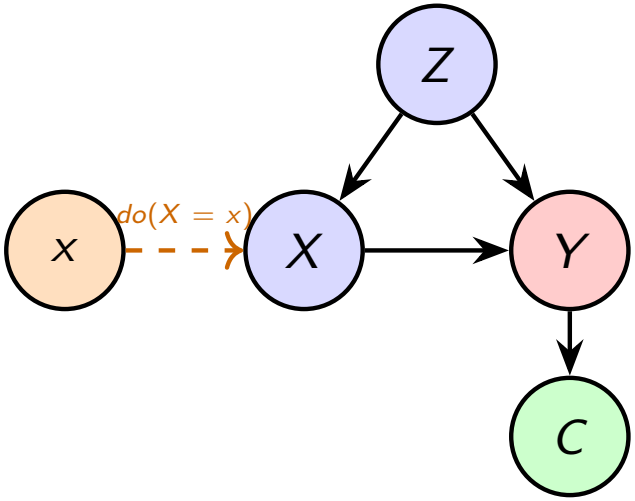
Z confounds X–Y

Manipulated world $do(X = x)$



arrow from Z is deleted

One DAG $\mathcal{G}$ encodes infinitely many distributions: each intervention $do(X_i = x)$ yields a new distribution $P^{do(X_i=x)}$ obtained by replacing the mechanism of X_i.

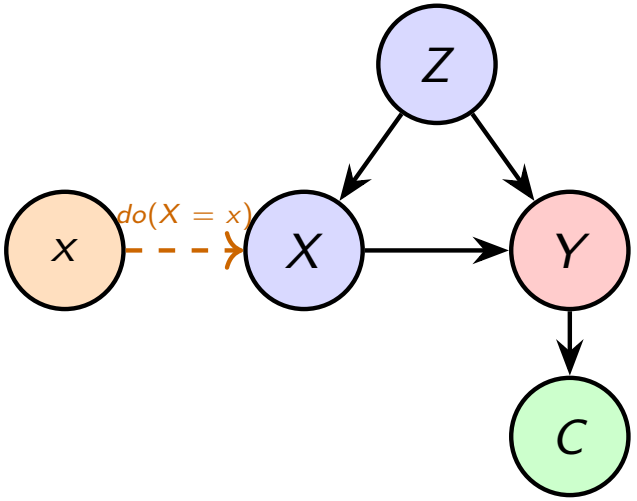


arrow $Z \longrightarrow X$ removed

One DAG $\mathcal{G}$ encodes infinitely many distributions: each intervention $do(X_i = x)$ yields a new distribution $P^{do(X_i=x)}$ obtained by replacing the mechanism of X_i.

Intervention class $\mathcal{P}(\mathcal{G})$:

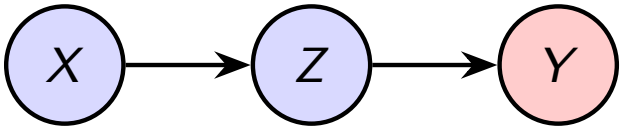
Fix a graph $\mathcal{G}$. The collection $\mathcal{P}(\mathcal{G}) = \{P^{do(\mathbf{X}_I=\mathbf{x})} : I \subseteq V, \mathbf{x}\}$ is the **interventional family** of $\mathcal{G}$.



arrow $Z \longrightarrow X$ removed

Chain

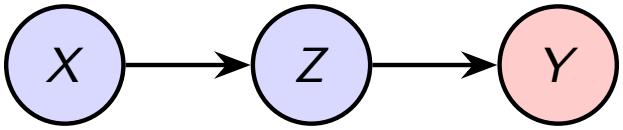
$X \to Z \to Y$



Conditioning on Z **blocks** the path.

Chain

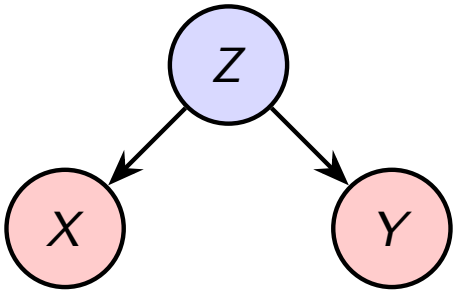
$X \rightarrow Z \rightarrow Y$



Conditioning on Z **blocks** the path.

Fork

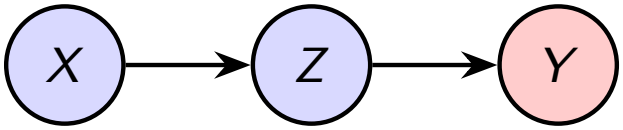
$X \leftarrow Z \rightarrow Y$



Conditioning on Z **blocks** the path.

Chain

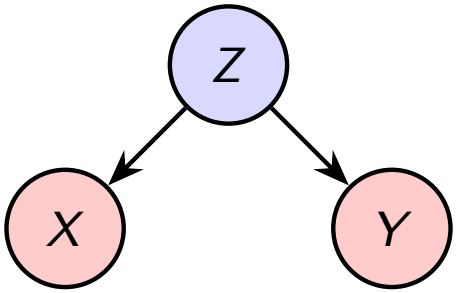
$X \rightarrow Z \rightarrow Y$



Conditioning on Z **blocks** the path.

Fork

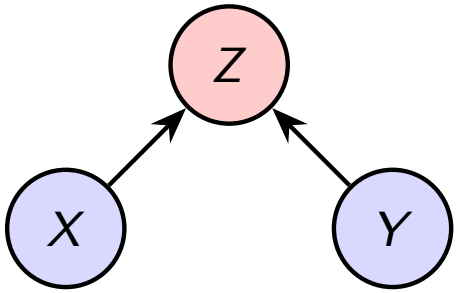
$X \leftarrow Z \rightarrow Y$



Conditioning on Z **blocks** the path.

Collider

$X \rightarrow Z \leftarrow Y$



Conditioning on Z **opens** the path.

	Causal parents $\mathrm{Pa}(Y)$	**Markov Blanket** $\mathrm{MB}(Y)$
No intervention (i.i.d data)	Lower	Higher
Intervention on *children*	✓	×
Intervention *non-MB* variables	✓	✓
Feature set size	Smaller	Larger
Learning from data	Harder	Easier

Practical guidance:

	Causal parents $\mathrm{Pa}(Y)$	**Markov Blanket** $\mathrm{MB}(Y)$
No intervention (i.i.d data)	Lower	Higher
Intervention on *children*	✓	×
Intervention *non-MB* variables	✓	✓
Feature set size	Smaller	Larger
Learning from data	Harder	Easier

Practical guidance:

- Use $\mathrm{MB}(Y)$ you expect test data distribution to be close to training data with distribution shifts arising outside the MB.

	Causal parents $\mathrm{Pa}(Y)$	**Markov Blanket** $\mathrm{MB}(Y)$
No intervention (i.i.d data)	Lower	Higher
Intervention on *children*	✓	×
Intervention *non-MB* variables	✓	✓
Feature set size	Smaller	Larger
Learning from data	Harder	Easier

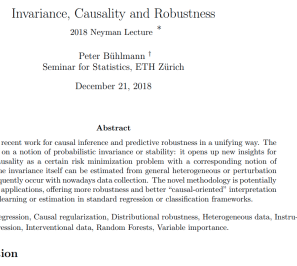
Practical guidance:

- Use $\mathrm{MB}(Y)$ you expect test data distribution to be close to training data with distribution shifts arising outside the MB.
- Use $\mathrm{Pa}(Y)$ when *strong* distribution shifts are expected with possible interventions on the children pf the target.

	Causal parents $\mathrm{Pa}(Y)$	**Markov Blanket** $\mathrm{MB}(Y)$
No intervention (i.i.d data)	Lower	Higher
Intervention on *children*	✓	×
Intervention *non-MB* variables	✓	✓
Feature set size	Smaller	Larger
Learning from data	Harder	Easier

Practical guidance:

- Use $\mathrm{MB}(Y)$ you expect test data distribution to be close to training data with distribution shifts arising outside the MB.
- Use $\mathrm{Pa}(Y)$ when *strong* distribution shifts are expected with possible interventions on the children pf the target.
- **The right choice depends on knowing *which* variables will be intervened on.**



Invariance, Causality and Robustness

2018 Neyman Lecture *

Peter Bühlmann †
Seminar for Statistics, ETH Zürich

December 21, 2018

Abstract

We discuss recent work for causal inference and predictive robustness in a unifying way. The key idea relies on a notion of probabilistic invariance or stability: it opens up new insights for formulating causality as a certain risk minimization problem with a corresponding notion of robustness. The invariance itself can be estimated from general heterogeneous or perturbation data which frequently occur with nowadays data collection. The novel methodology is potentially useful in many applications, offering more robustness and better "causal-oriented" interpretation than machine learning or estimation in standard regression or classification frameworks.

Keywords: Anchor regression, Causal regularization, Distributional robustness, Heterogeneous data, Instrumental variables regression, Interventional data, Random Forests, Variable importance.

1 Introduction

ROYAL STATISTICAL SOCIETY
DATA | EVIDENCE | DECISIONS
Journal of the Royal Statistical Society
Statistical Methodology
Series B

J. R. Statist. Soc. B (2016)
78, *Part* 5, *pp.* 947–1012

Causal inference by using invariant prediction: identification and confidence intervals

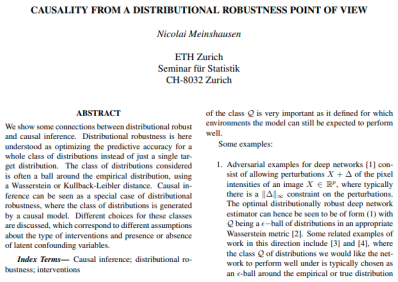
Jonas Peters

Max Planck Institute for Intelligent Systems, Tübingen, Germany, and Eidgenössiche Technische Hochschule Zürich, Switzerland

and Peter Bühlmann and Nicolai Meinshausen

Eidgenössiche Technische Hochschule Zürich, Switzerland

[*Read before* The Royal Statistical Society *at a meeting organized by the* Research Section *on Wednesday, May 11th, 2016,* Professor C. Leng *in the Chair*]



CAUSALITY FROM A DISTRIBUTIONAL ROBUSTNESS POINT OF VIEW

Nicolai Meinshausen

ETH Zurich
Seminar für Statistik
CH-8032 Zurich

ABSTRACT

We show some connections between distributional robust and causal inference. Distributional robustness is here understood as optimizing the predictive accuracy for a whole class of distributions instead of just a single target distribution. The class of distributions considered is often a ball around the empirical distribution, using a Wasserstein or Kullback-Leibler distance. Causal inference can be seen as a special case of distributional robustness, where the class of distributions is generated by a causal model. Different choices for these classes are discussed, which correspond to different assumptions about the type of interventions and presence or absence of latent confounding variables.

***Index Terms*—** Causal inference; distributional robustness; interventions

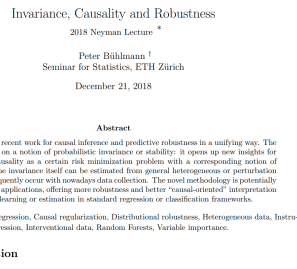
of the class $\mathcal{Q}$ is very important as it defined for which environments the model can still be expected to perform well.

Some examples:

1. Adversarial examples for deep networks [1] consist of allowing perturbations $X + \Delta$ of the pixel intensities of an image $X \in \mathbb{R}^p$, where typically there is a $\|\Delta\|_\infty$ constraint on the perturbations. The optimal distributionally robust deep network estimator can hence be seen to be of form (1) with $\mathcal{Q}$ being a $\epsilon-$ball of distributions in an appropriate Wasserstein metric [2]. Some related examples of work in this direction include [3] and [4], where the class $\mathcal{Q}$ of distributions we would like the network to perform well under is typically chosen as an ϵ-ball around the empirical or true distribution

Important Papers

- **Invariance, Causality and Robustness**
  Meinshausen (2018)



Invariance, Causality and Robustness

2018 Neyman Lecture *

Peter Bühlmann †
Seminar for Statistics, ETH Zürich

December 21, 2018

Abstract

We discuss recent work for causal inference and predictive robustness in a unifying way. The key idea relies on a notion of probabilistic invariance or stability: it opens up new insights for formulating causality as a certain risk minimization problem with a corresponding notion of robustness. The invariance itself can be estimated from general heterogeneous or perturbation data which frequently occur with nowadays data collection. The novel methodology is potentially useful in many applications, offering more robustness and better "causal-oriented" interpretation than machine learning or estimation in standard regression or classification frameworks.

Keywords: Anchor regression, Causal regularization, Distributional robustness, Heterogeneous data, Instrumental variables regression, Interventional data, Random Forests, Variable importance.

1 Introduction

ROYAL STATISTICAL SOCIETY
DATA | EVIDENCE | DECISIONS
Journal of the Royal Statistical Society
Statistical Methodology
Series B

J. R. Statist. Soc. B (2016)
78, *Part* 5, *pp.* 947–1012

Causal inference by using invariant prediction: identification and confidence intervals

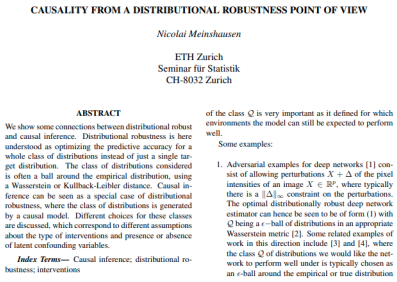
Jonas Peters

Max Planck Institute for Intelligent Systems, Tübingen, Germany, and Eidgenössiche Technische Hochschule Zürich, Switzerland

and Peter Bühlmann and Nicolai Meinshausen

Eidgenössiche Technische Hochschule Zürich, Switzerland

[*Read before* The Royal Statistical Society *at a meeting organized by the* Research Section *on Wednesday, May 11th, 2016,* Professor C. Leng *in the Chair*]



CAUSALITY FROM A DISTRIBUTIONAL ROBUSTNESS POINT OF VIEW

Nicolai Meinshausen

ETH Zurich
Seminar für Statistik
CH-8032 Zurich

ABSTRACT

We show some connections between distributional robust and causal inference. Distributional robustness is here understood as optimizing the predictive accuracy for a whole class of distributions instead of just a single target distribution. The class of distributions considered is often a ball around the empirical distribution, using a Wasserstein or Kullback-Leibler distance. Causal inference can be seen as a special case of distributional robustness, where the class of distributions is generated by a causal model. Different choices for these classes are discussed, which correspond to different assumptions about the type of interventions and presence or absence of latent confounding variables.

***Index Terms*—** Causal inference; distributional robustness; interventions

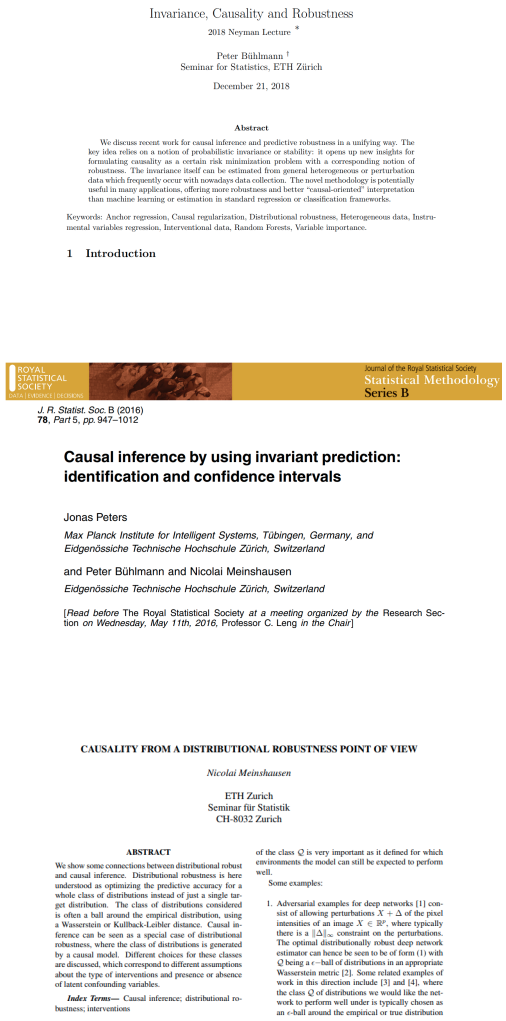
of the class $\mathcal{Q}$ is very important as it defined for which environments the model can still be expected to perform well.

Some examples:

1. Adversarial examples for deep networks [1] consist of allowing perturbations $X + \Delta$ of the pixel intensities of an image $X \in \mathbb{R}^p$, where typically there is a $\|\Delta\|_\infty$ constraint on the perturbations. The optimal distributionally robust deep network estimator can hence be seen to be of form (1) with $\mathcal{Q}$ being a $\epsilon-$ball of distributions in an appropriate Wasserstein metric [2]. Some related examples of work in this direction include [3] and [4], where the class $\mathcal{Q}$ of distributions we would like the network to perform well under is typically chosen as an ϵ-ball around the empirical or true distribution

Important Papers

- **Invariance, Causality and Robustness**
  Meinshausen (2018)
- **Causal inference by using invariant prediction**
  Peters et al. (2016)



Invariance, Causality and Robustness

2018 Neyman Lecture *

Peter Bühlmann †
Seminar for Statistics, ETH Zürich

December 21, 2018

Abstract

We discuss recent work for causal inference and predictive robustness in a unifying way. The key idea relies on a notion of probabilistic invariance or stability: it opens up new insights for formulating causality as a certain risk minimization problem with a corresponding notion of robustness. The invariance itself can be estimated from general heterogeneous or perturbation data which frequently occur with nowadays data collection. The novel methodology is potentially useful in many applications, offering more robustness and better "causal-oriented" interpretation than machine learning or estimation in standard regression or classification frameworks.

Keywords: Anchor regression, Causal regularization, Distributional robustness, Heterogeneous data, Instrumental variables regression, Interventional data, Random Forests, Variable importance.

1 Introduction

ROYAL STATISTICAL SOCIETY

Journal of the Royal Statistical Society
Statistical Methodology
Series B

J. R. Statist. Soc. B (2016)
78, *Part* 5, *pp.* 947–1012

Causal inference by using invariant prediction: identification and confidence intervals

Jonas Peters
Max Planck Institute for Intelligent Systems, Tübingen, Germany, and Eidgenössiche Technische Hochschule Zürich, Switzerland

and Peter Bühlmann and Nicolai Meinshausen
Eidgenössiche Technische Hochschule Zürich, Switzerland

[*Read before* The Royal Statistical Society *at a meeting organized by the* Research Section *on Wednesday, May 11th, 2016,* Professor C. Leng *in the Chair*]

CAUSALITY FROM A DISTRIBUTIONAL ROBUSTNESS POINT OF VIEW

Nicolai Meinshausen

ETH Zurich
Seminar für Statistik
CH-8032 Zurich

ABSTRACT

We show some connections between distributional robust and causal inference. Distributional robustness is here understood as optimizing the predictive accuracy for a whole class of distributions instead of just a single target distribution. The class of distributions considered is often a ball around the empirical distribution, using a Wasserstein or Kullback-Leibler distance. Causal inference can be seen as a special case of distributional robustness, where the class of distributions is generated by a causal model. Different choices for these classes are discussed, which correspond to different assumptions about the type of interventions and presence or absence of latent confounding variables.

***Index Terms*—** Causal inference; distributional robustness; interventions

of the class $\mathcal{Q}$ is very important as it defined for which environments the model can still be expected to perform well.

Some examples:

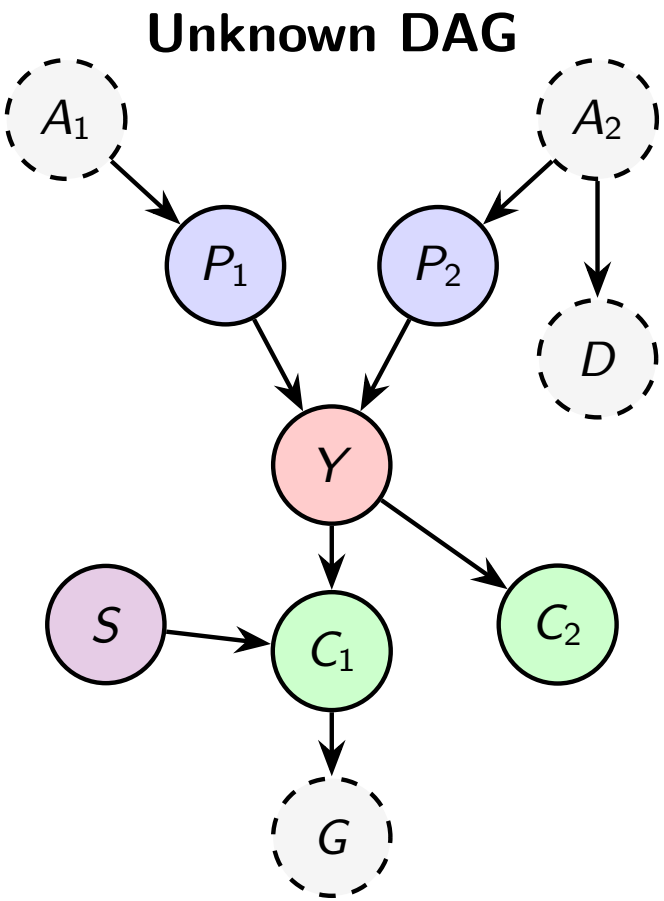
1. Adversarial examples for deep networks [1] consist of allowing perturbations $X + \Delta$ of the pixel intensities of an image $X \in \mathbb{R}^p$, where typically there is a $\|\Delta\|_\infty$ constraint on the perturbations. The optimal distributionally robust deep network estimator can hence be seen to be of form (1) with $\mathcal{Q}$ being a $\epsilon-$ball of distributions in an appropriate Wasserstein metric [2]. Some related examples of work in this direction include [3] and [4], where the class $\mathcal{Q}$ of distributions we would like the network to perform well under is typically chosen as an ϵ-ball around the empirical or true distribution

Important Papers

- **Invariance, Causality and Robustness**
  Meinshausen (2018)
- **Causal inference by using invariant prediction**
  Peters et al. (2016)
- **Causality from a distributional robustness point of view**
  Bühlmann (2020)

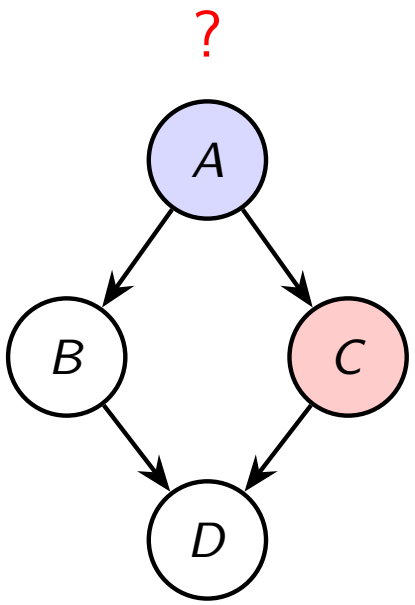
3 Outline

1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion



	A_1	A_2	P_1	P_2	C_1	C_2	S	G	D	Y
0	0.8	-0.2	1.1	0.5	-0.7	0.3	-1.1	0.4	0.9	0.7
1	-0.4	1.0	0.4	-1.2	0.2	-0.5	0.6	-0.8	0.1	-0.3
2	1.1	-0.6	0.7	0.9	-0.3	0.1	-0.3	0.5	-0.6	0.9
3	-0.8	0.1	-0.5	1.3	0.5	-0.9	0.1	-0.2	0.4	-0.7
4	0.3	0.6	-1.1	-0.4	1.0	0.6	-0.7	0.3	-0.2	0.2
5	-0.1	-1.2	0.5	0.8	-0.2	1.0	0.9	-0.5	0.3	0.6
6	0.6	-0.7	0.9	0.2	-0.4	0.5	-0.6	0.7	-0.1	0.5
7	-1.0	0.4	-0.3	1.1	0.6	-0.2	0.2	-0.6	0.5	-0.4
8	0.2	-0.9	0.8	0.7	-0.1	0.4	-0.8	0.1	-0.7	0.6
9	-0.5	0.3	-0.7	1.0	0.3	-0.6	0.4	-0.3	0.2	-0.5
10	0.9	-0.4	1.2	0.3	-0.5	0.8	-0.2	0.6	-0.4	0.8
11	-0.6	0.8	0.1	-1.0	0.4	-0.3	0.7	-0.7	0.0	-0.2
12	0.4	-0.1	0.6	0.6	-0.2	0.2	-0.4	0.2	-0.3	0.4
13	-0.9	0.5	-0.4	1.2	0.7	-0.8	0.3	-0.4	0.6	-0.6
14	0.1	-0.8	0.3	0.9	-0.6	0.7	-0.5	0.3	-0.5	0.3
15	-0.2	0.7	0.2	-0.9	0.1	-0.4	0.8	-0.1	0.4	-0.1

Causal discovery aims to recover $\mathcal{G}$ from data.



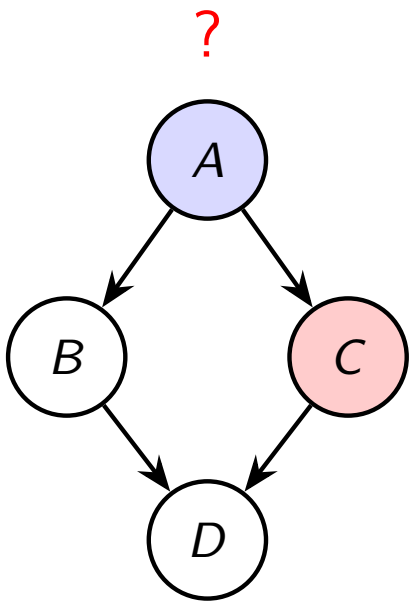
Which arrows can data reveal?

3 Different approaches

Causal discovery aims to recover $\mathcal{G}$ from data.

Three families of methods:

1. **Constraint-based**
   Use conditional independence tests to rule out or confirm edges (PC algorithm).



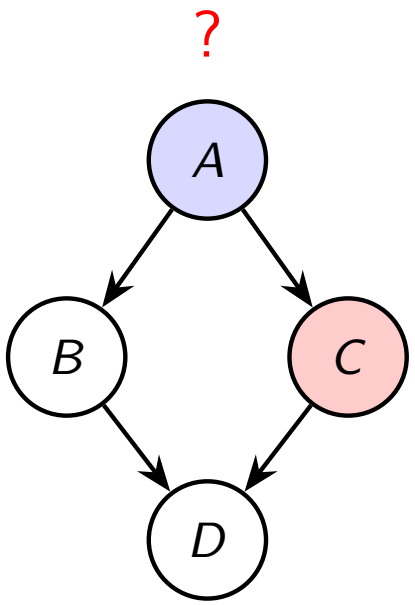
Which arrows can data reveal?

3 Different approaches

Causal discovery aims to recover $\mathcal{G}$ from data.

Three families of methods:

1. **Constraint-based**
   Use conditional independence tests to rule out or confirm edges (PC algorithm).
2. **Score-based**
   Search over graphs, score each by how well it fits the data (GES algorithm).



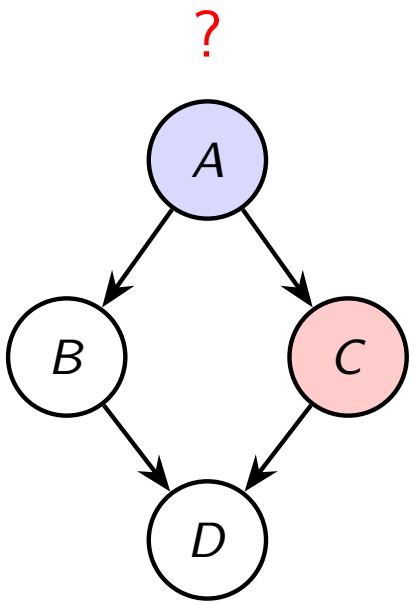
Which arrows can data reveal?

3 Different approaches

Causal discovery aims to recover $\mathcal{G}$ from data.

Three families of methods:

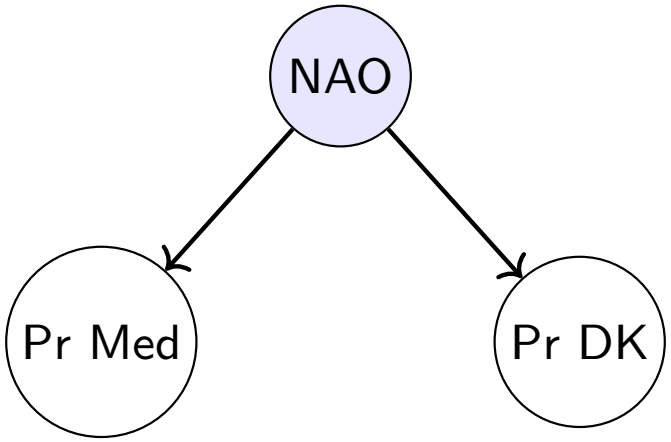
1. **Constraint-based**
   Use conditional independence tests to rule out or confirm edges (PC algorithm).
2. **Score-based**
   Search over graphs, score each by how well it fits the data (GES algorithm).
3. **Functional / asymmetry-based**
   Exploit the mathematical asymmetry between cause and effect (Additive Noise Models).



Which arrows can data reveal?

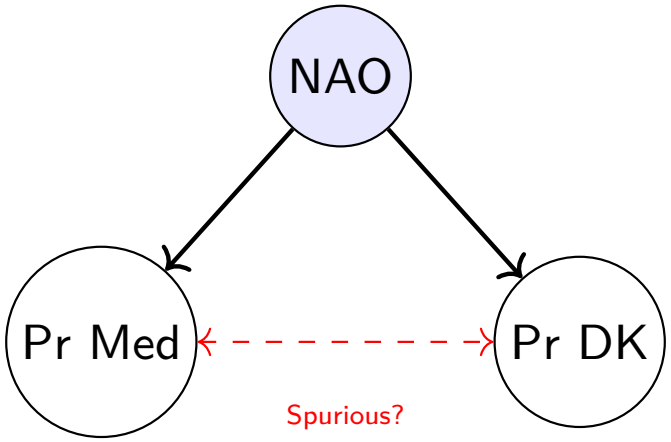
Goal: finding (d) separation in the graph to be able to recover it.

1. **Intuition**
   - $X \perp Y \mid Z$: Once we account for the state of Z, does X tell us anything new about Y?



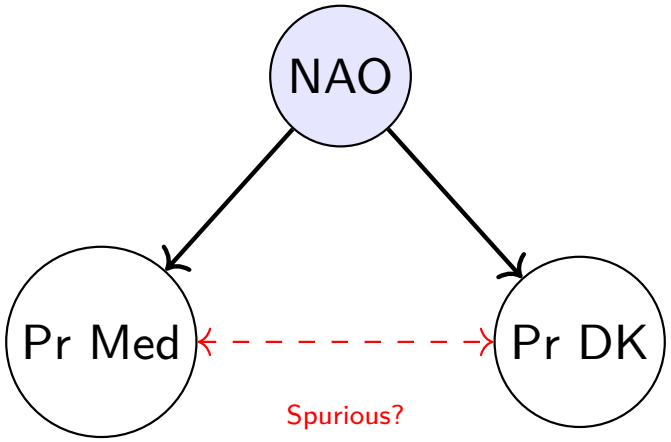
Goal: finding (d) separation in the graph to be able to recover it.

1. **Intuition**
   - $X \perp Y \mid Z$: Once we account for the state of Z, does X tell us anything new about Y?
2. **Linear CI**
   - **Partial Correlation:** Correlation between residuals after regressing out Z (also known as **fisher-z** test).



Goal: finding (d) separation in the graph to be able to recover it.

1. **Intuition**
   - $X \perp Y \mid Z$: Once we account for the state of Z, does X tell us anything new about Y?
2. **Linear CI**
   - **Partial Correlation:** Correlation between residuals after regressing out Z (also known as **fisher-z** test).
3. **Nonlinear CI**
   - **KCI:** Detects dependencies that linear tests miss.
   - Based on kernel dependence measure

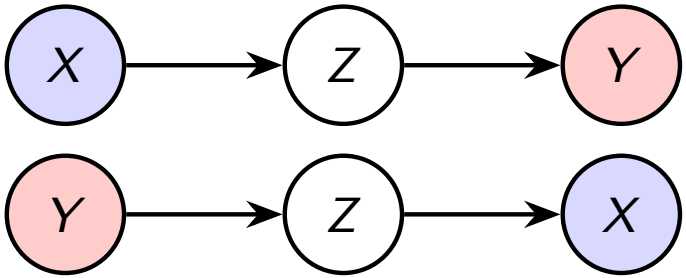


Fundamental limit: observational data[1] can only identify the **Markov Equivalence Class (MEC)** — the set of DAGs that encode the *same* conditional independencies.

[1] Without further functional assumptions.

Fundamental limit: observational data[1] can only identify the **Markov Equivalence Class (MEC)** — the set of DAGs that encode the *same* conditional independencies.

What *is not* identifiable:



Both imply $X \perp\!\!\!\perp Y \mid Z$; data alone cannot choose.

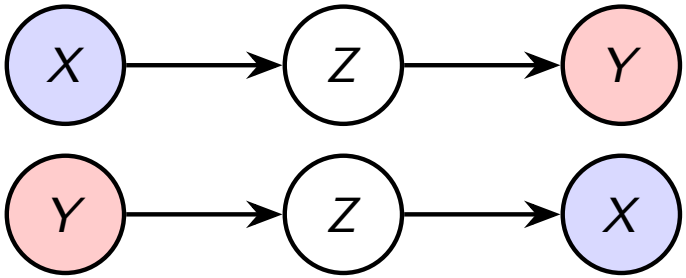
[1] Without further functional assumptions.

Fundamental limit: observational data[1] can only identify the **Markov Equivalence Class (MEC)** — the set of DAGs that encode the *same* conditional independencies.

What *is* identifiable:

- The **skeleton** (which pairs are connected)

What *is not* identifiable:



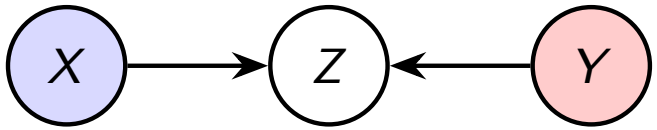
Both imply $X \perp\!\!\!\perp Y \mid Z$; data alone cannot choose.

[1] Without further functional assumptions.

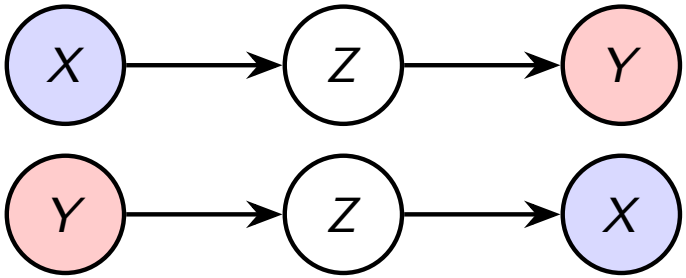
Fundamental limit: observational data[1] can only identify the **Markov Equivalence Class (MEC)** — the set of DAGs that encode the *same* conditional independencies.

What *is* identifiable:

- The **skeleton** (which pairs are connected)
- **V-structures**: $X \to Z \leftarrow Y$ (colliders) are uniquely identifiable



What *is not* identifiable:



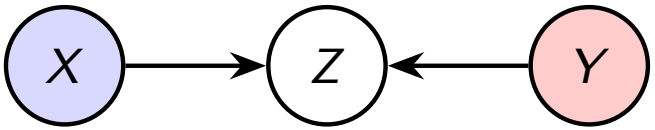
Both imply $X \perp\!\!\!\perp Y \mid Z$; data alone cannot choose.

[1] Without further functional assumptions.

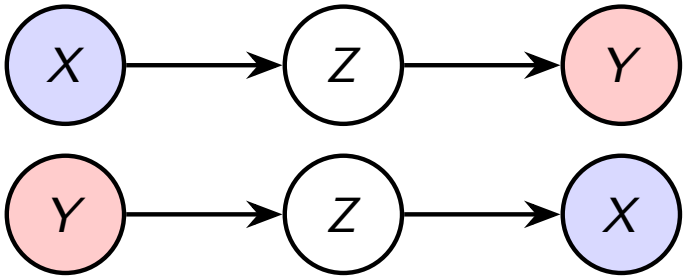
Fundamental limit: observational data[1] can only identify the **Markov Equivalence Class (MEC)** — the set of DAGs that encode the *same* conditional independencies.

What *is* identifiable:

- The **skeleton** (which pairs are connected)
- **V-structures**: $X \to Z \leftarrow Y$ (colliders) are uniquely identifiable
- The output is a **CPDAG** (Completed Partially Directed Acyclic Graph) — some edges remain undirected

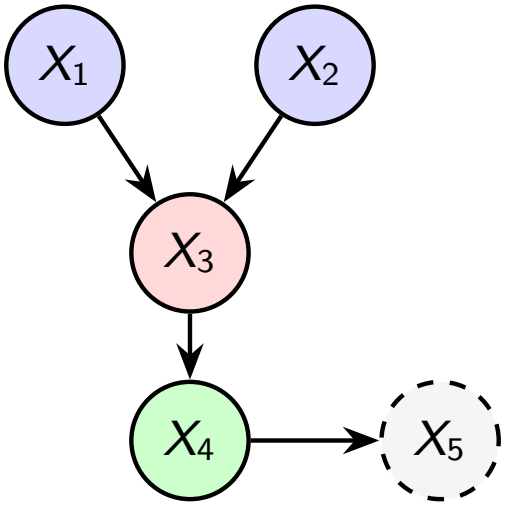


What *is not* identifiable:

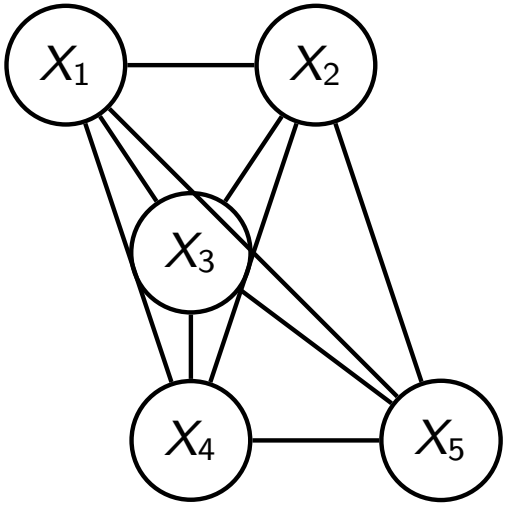


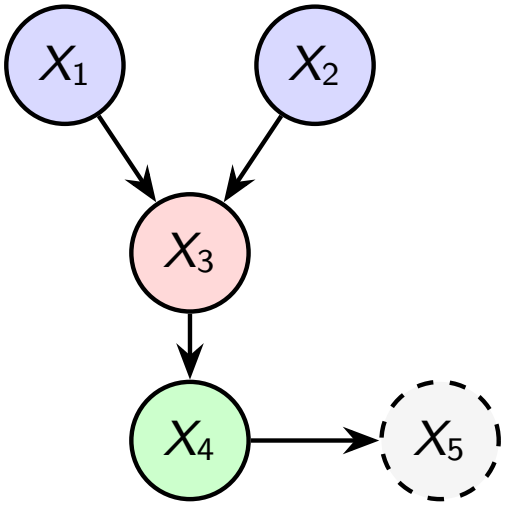
Both imply $X \perp\!\!\!\perp Y \mid Z$; data alone cannot choose.

[1] Without further functional assumptions.

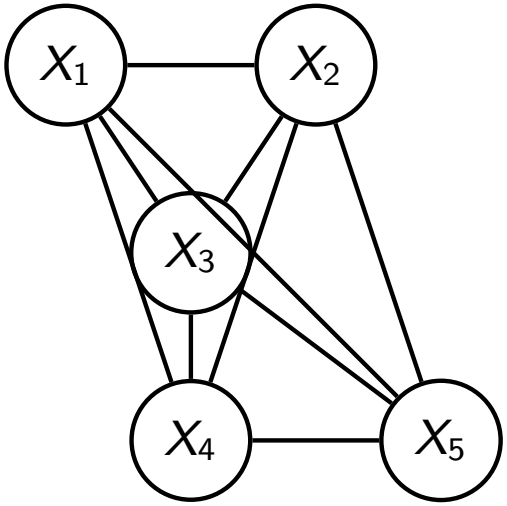

X_1
X_2
X_3
X_4
X_5

Step 1: Full graph

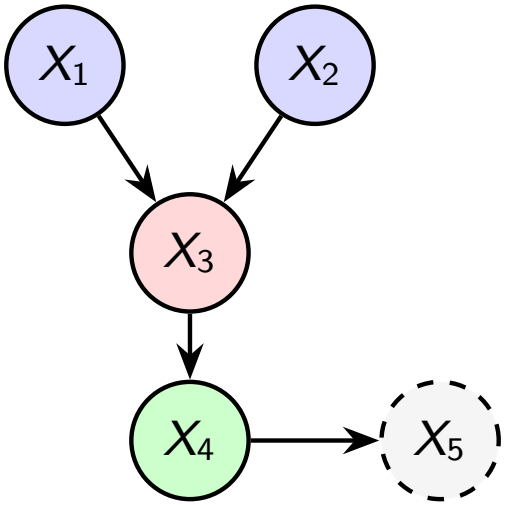




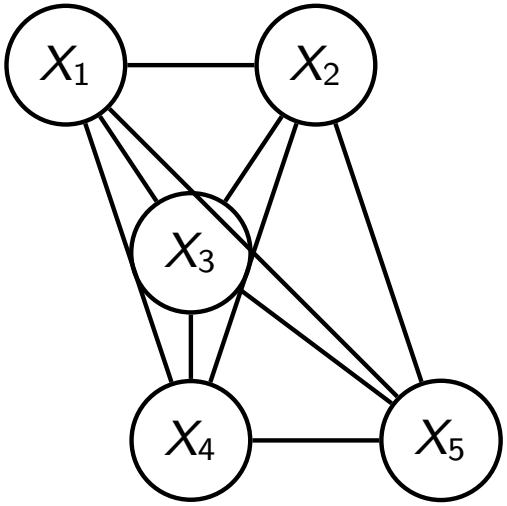
Step 1: Full graph



Test $X_i \perp X_j | S$

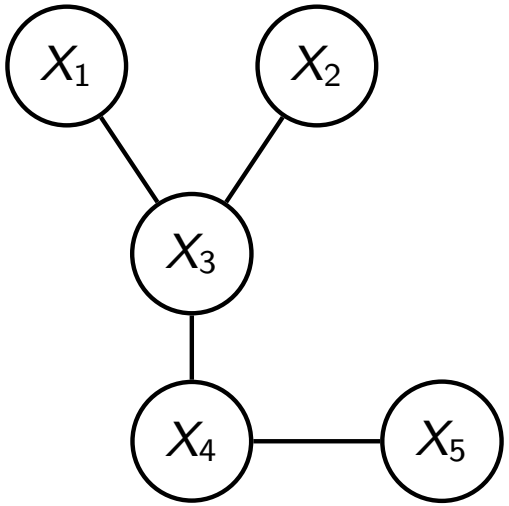


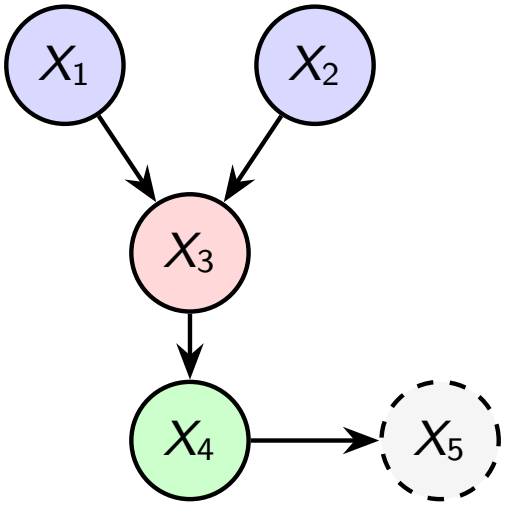
Step 1: Full graph



Test $X_i \perp X_j | S$

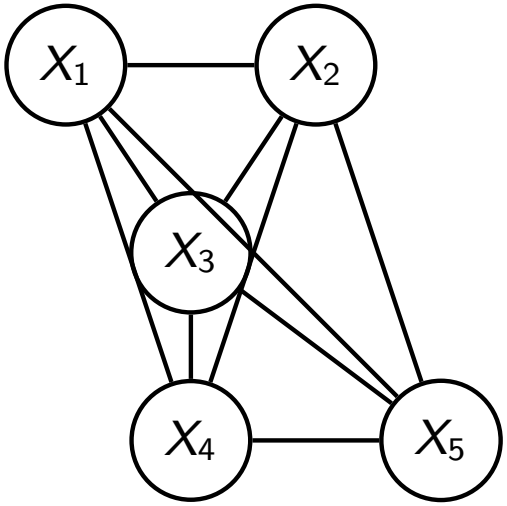
Step 2: Skeleton





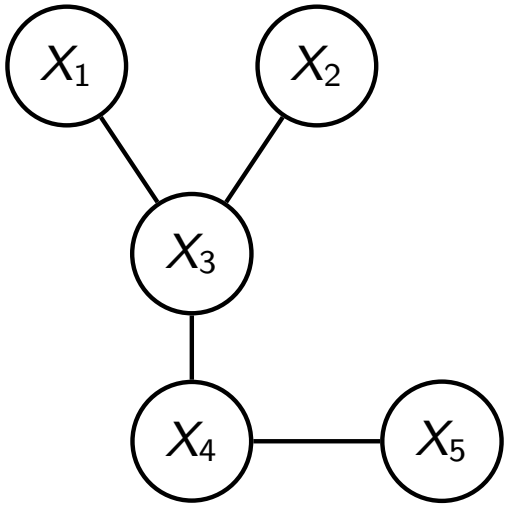
3 PC algorithm

Step 1: Full graph

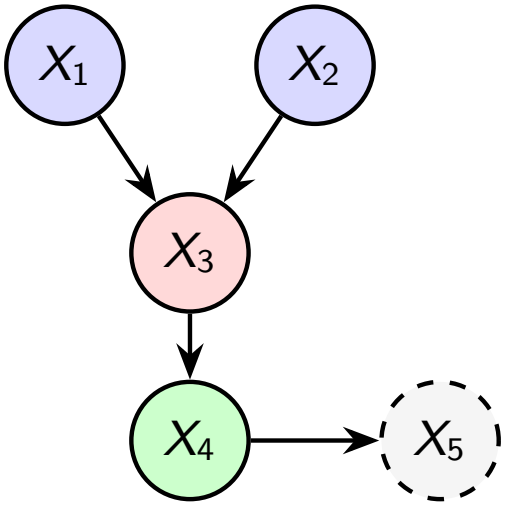


Test $X_i \perp X_j | S$

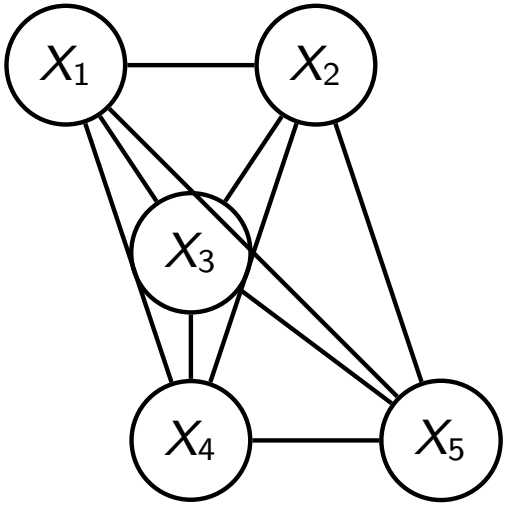
Step 2: Skeleton



V-structures/ Meek rule

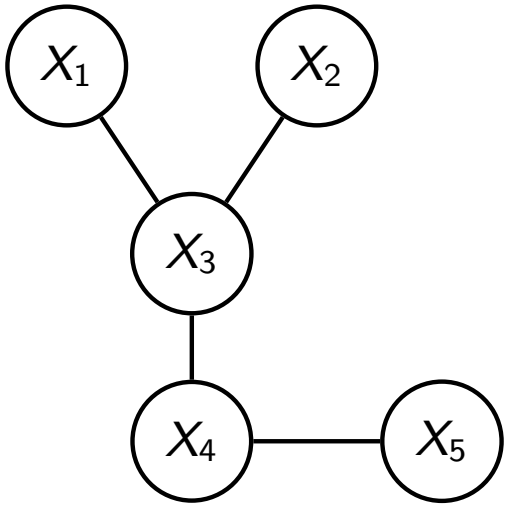


Step 1: Full graph



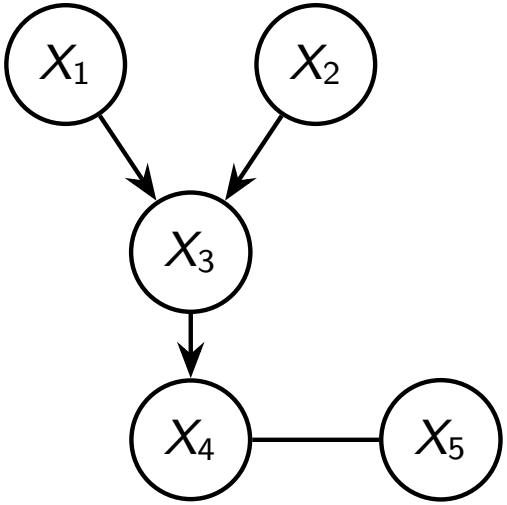
Test $X_i \perp X_j | S$

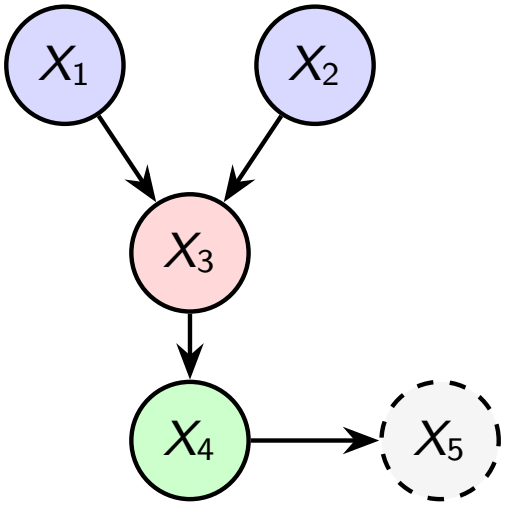
Step 2: Skeleton



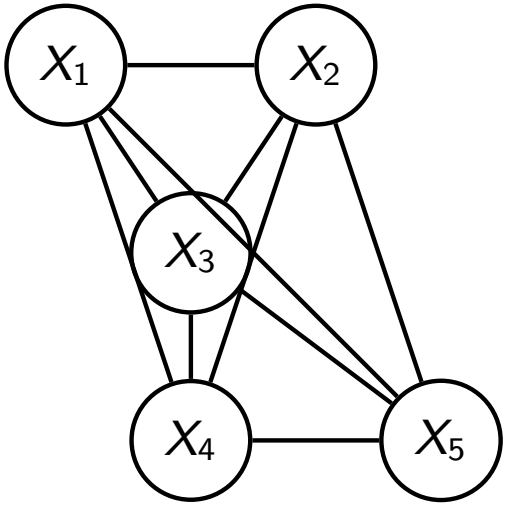
V-structures/ Meek rule

Steps 3–4: CPDAG



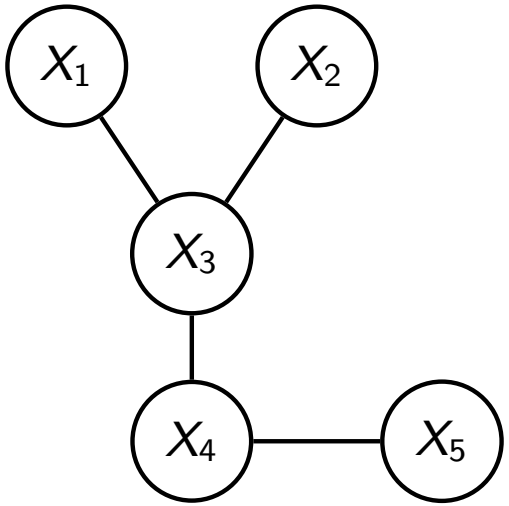


Step 1: Full graph



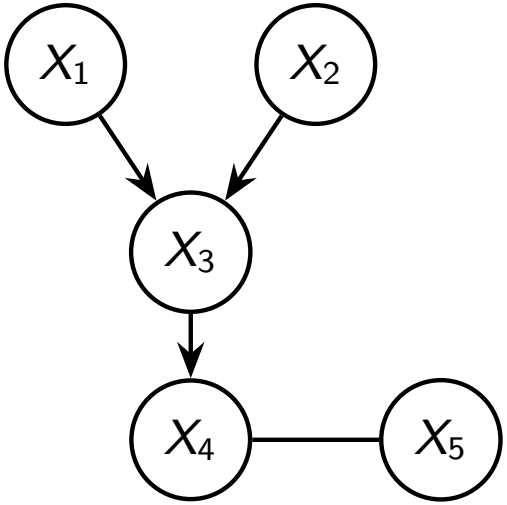
Test $X_i \perp X_j | S$

Step 2: Skeleton

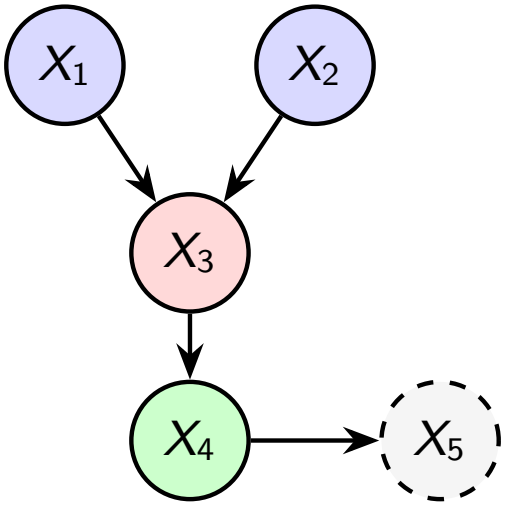


V-structures/ Meek rule

Steps 3–4: CPDAG



Output: CPDAG



3 The GES Algorithm (Score-Based)

Idea: search over possible graphs and *score* each one by how well it fits the data, penalising complexity.

3 The GES Algorithm (Score-Based)

Idea: search over possible graphs and *score* each one by how well it fits the data, penalising complexity.

Score (BIC): $\mathrm{BIC} = \log \hat{L} - \frac{k}{2} \log n$ (goodness of fit minus model complexity penalty)

3 The GES Algorithm (Score-Based)

Idea: search over possible graphs and *score* each one by how well it fits the data, penalising complexity.

Score (BIC): $\text{BIC} = \log \hat{L} - \frac{k}{2} \log n$ (goodness of fit minus model complexity penalty)

1. **Forward phase:** start from empty graph, greedily add the edge that most improves BIC, until no improvement.

3 The GES Algorithm (Score-Based)

Idea: search over possible graphs and *score* each one by how well it fits the data, penalising complexity.

Score (BIC): $\mathrm{BIC} = \log \hat{L} - \frac{k}{2} \log n$ (goodness of fit minus model complexity penalty)

1. **Forward phase:** start from empty graph, greedily add the edge that most improves BIC, until no improvement.
2. **Backward phase:** greedily remove edges, stop when no removal helps.

3 The GES Algorithm (Score-Based)

Idea: search over possible graphs and *score* each one by how well it fits the data, penalising complexity.

Score (BIC): $\text{BIC} = \log \hat{L} - \frac{k}{2} \log n$ (goodness of fit minus model complexity penalty)

1. **Forward phase:** start from empty graph, greedily add the edge that most improves BIC, until no improvement.
2. **Backward phase:** greedily remove edges, stop when no removal helps.

Why is GES useful?

- A single wrong independence test in PC can cascade through the whole graph.

3 The GES Algorithm (Score-Based)

Idea: search over possible graphs and *score* each one by how well it fits the data, penalising complexity.

Score (BIC): $\mathrm{BIC} = \log \hat{L} - \frac{k}{2} \log n$ (goodness of fit minus model complexity penalty)

1. **Forward phase:** start from empty graph, greedily add the edge that most improves BIC, until no improvement.
2. **Backward phase:** greedily remove edges, stop when no removal helps.

Why is GES useful?

- A single wrong independence test in PC can cascade through the whole graph.
- GES is **more robust to small samples** because each edge decision uses a global score, not a single test.

3 The GES Algorithm (Score-Based)

Idea: search over possible graphs and *score* each one by how well it fits the data, penalising complexity.

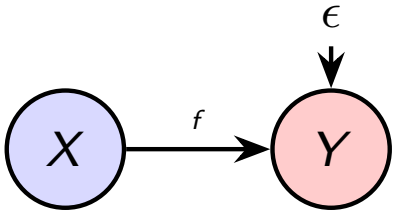
Score (BIC): $\text{BIC} = \log \hat{L} - \frac{k}{2} \log n$ (goodness of fit minus model complexity penalty)

1. **Forward phase:** start from empty graph, greedily add the edge that most improves BIC, until no improvement.
2. **Backward phase:** greedily remove edges, stop when no removal helps.

Why is GES useful?

- A single wrong independence test in PC can cascade through the whole graph.
- GES is **more robust to small samples** because each edge decision uses a global score, not a single test.
- Same output as PC: a CPDAG (equivalence class).

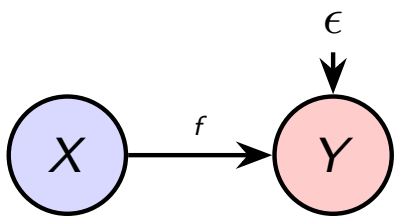
Problem: PC and GES cannot distinguish a chain $X \to Y$ from $Y \to X$ (same MEC).



Problem: PC and GES cannot distinguish a chain $X \to Y$ from $Y \to X$ (same MEC).

In the *true* causal direction, the noise is independent of the cause.

$$Y = f(X) + \epsilon, \qquad \epsilon \perp\!\!\!\perp X$$



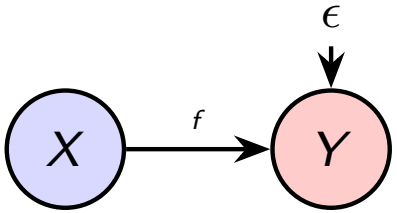
Problem: PC and GES cannot distinguish a chain $X \to Y$ from $Y \to X$ (same MEC).

In the *true* causal direction, the noise is independent of the cause.

$$Y = f(X) + \epsilon, \qquad \epsilon \perp\!\!\!\perp X$$

Test:

1 Fit $\hat{f}$ by regression Y on X.



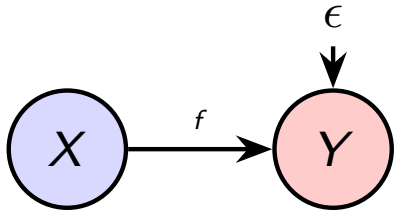
Problem: PC and GES cannot distinguish a chain $X \to Y$ from $Y \to X$ (same MEC).

In the *true* causal direction, the noise is independent of the cause.

$$Y = f(X) + \epsilon, \qquad \epsilon \perp\!\!\!\perp X$$

Test:

1. Fit $\hat{f}$ by regression Y on X.
2. Check whether residuals $\hat{\epsilon} = Y - \hat{f}(X)$ are independent of X.



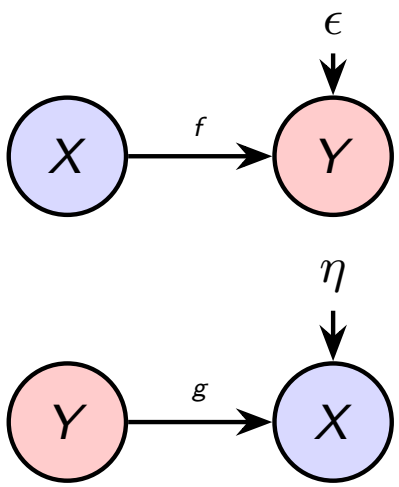
Problem: PC and GES cannot distinguish a chain $X \to Y$ from $Y \to X$ (same MEC).

In the *true* causal direction, the noise is independent of the cause.

$$Y = f(X) + \epsilon, \qquad \epsilon \perp\!\!\!\perp X$$

Test:

1. Fit $\hat{f}$ by regression Y on X.
2. Check whether residuals $\hat{\epsilon} = Y - \hat{f}(X)$ are independent of X.
3. Try the reverse direction: $X = g(Y) + \eta$.



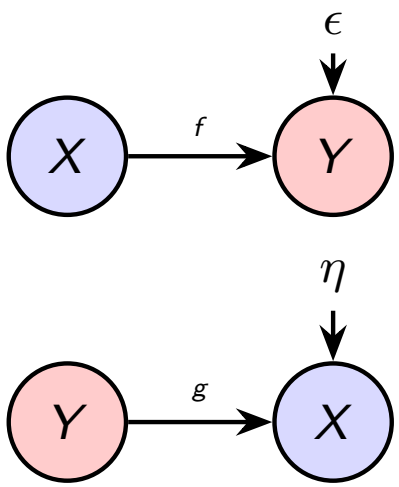
Problem: PC and GES cannot distinguish a chain $X \to Y$ from $Y \to X$ (same MEC).

In the *true* causal direction, the noise is independent of the cause.

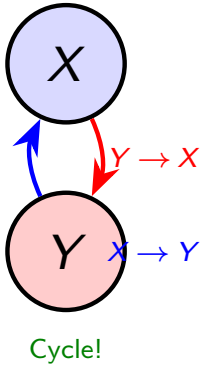
$$Y = f(X) + \epsilon, \qquad \epsilon \perp\!\!\!\perp X$$

Test:

1. Fit $\hat{f}$ by regression Y on X.
2. Check whether residuals $\hat{\epsilon} = Y - \hat{f}(X)$ are independent of X.
3. Try the reverse direction: $X = g(Y) + \eta$.
4. Only *one* direction passes the independence test.

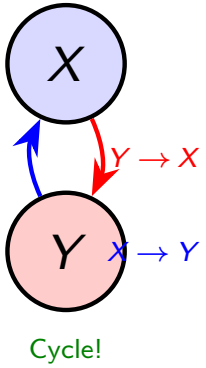


Cyclic graph (instantaneous)



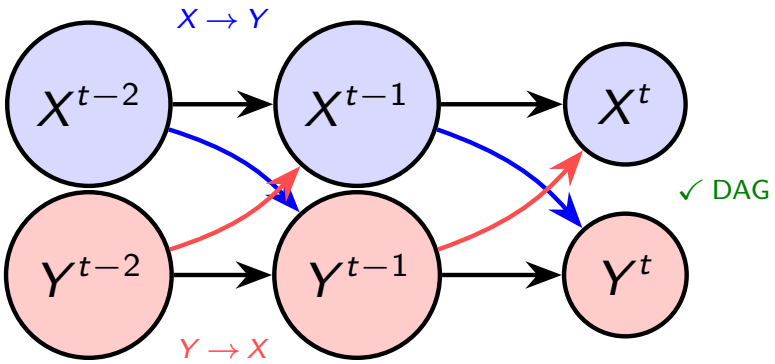
Apparent feedback $X \leftrightarrows Y$ forms a *cycle*, which violates standard DAG assumptions.

Cyclic graph (instantaneous)



Apparent feedback $X \leftrightarrows Y$ forms a *cycle*, which violates standard DAG assumptions.

Time-unfolded DAG



Time-lagged nodes transform the instantaneous cycle into a DAG, allowing PC-style CI tests to work.

Feature	PC (Constraint)	GES (Score)	ANM (Functional)
Category	Constraint based	Score based	Functional

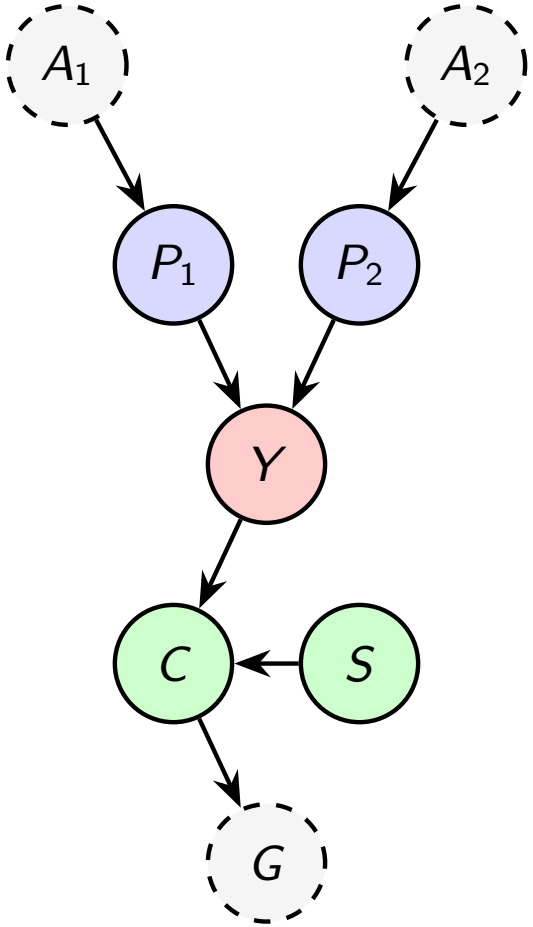
3 PC, GES and ANM

Feature	PC (Constraint)	GES (Score)	ANM (Functional)
Category	Constraint based	Score based	Functional
Output	CPDAG	CPDAG	Fully Directed

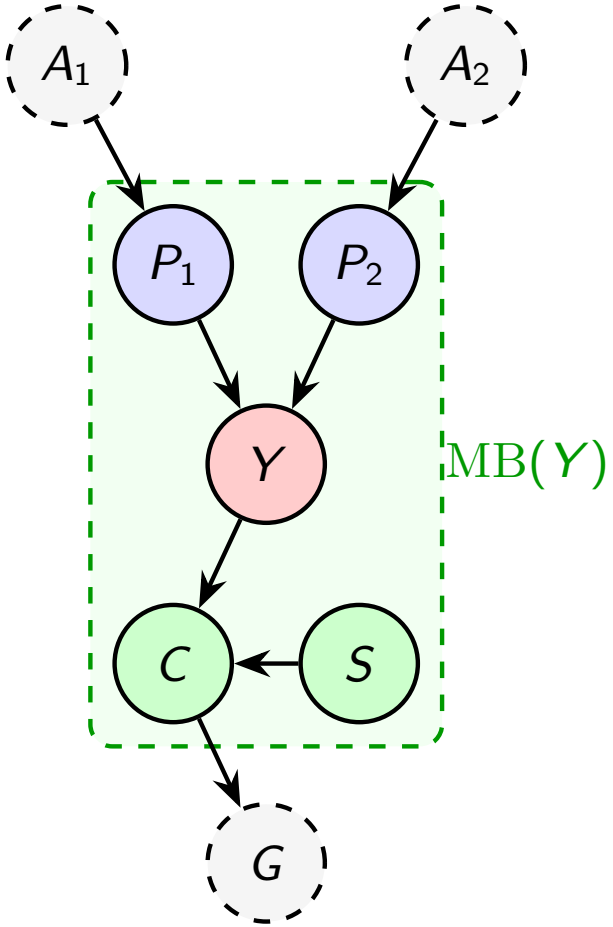
3 PC, GES and ANM

Feature	PC (Constraint)	GES (Score)	ANM (Functional)
Category	Constraint based	Score based	Functional
Output	CPDAG	CPDAG	Fully Directed
Assumes	Faithfulness/Markov	Faithfulness/Markov	Non-Gaussian/Linear

Full causal discovery is often overkill.
For robust prediction, we only need the *local* structure around Y: its parents, children, and spouses.



Full causal discovery is often overkill.
For robust prediction, we only need the *local* structure around Y: its parents, children, and spouses.



Complexity Note:
Full discovery is often **NP-hard**. Local MB methods are typically $O(p^2)$ or even $O(p)$ in the number of conditional independence (CI) tests.

Algorithm	Approach	Output	Strength

Complexity Note:
Full discovery is often **NP-hard**. Local MB methods are typically $O(p^2)$ or even $O(p)$ in the number of conditional independence (CI) tests.

Algorithm	Approach	Output	Strength
HITON-MB	CI-based	$\mathrm{MB}(Y)$	High-dimensional data

Complexity Note:
Full discovery is often **NP-hard**. Local MB methods are typically $O(p^2)$ or even $O(p)$ in the number of conditional independence (CI) tests.

Algorithm	Approach	Output	Strength
HITON-MB	CI-based	$\mathrm{MB}(Y)$	High-dimensional data
IAMB	CI-based	$\mathrm{MB}(Y)$	Converges fast

Complexity Note:
Full discovery is often **NP-hard**. Local MB methods are typically $O(p^2)$ or even $O(p)$ in the number of conditional independence (CI) tests.

Algorithm	Approach	Output	Strength
HITON-MB	CI-based	$\mathrm{MB}(Y)$	High-dimensional data
IAMB	CI-based	$\mathrm{MB}(Y)$	Converges fast
GES-MB	Score-based	$\mathrm{MB}(Y)$	Small sample size

Complexity Note:
Full discovery is often **NP-hard**. Local MB methods are typically $O(p^2)$ or even $O(p)$ in the number of conditional independence (CI) tests.

Algorithm	Approach	Output	Strength
HITON-MB	CI-based	$\mathrm{MB}(Y)$	High-dimensional data
IAMB	CI-based	$\mathrm{MB}(Y)$	Converges fast
GES-MB	Score-based	$\mathrm{MB}(Y)$	Small sample size
linGAM	Functional	$\mathrm{Pa}(Y)$	Directs causal edges

Complexity Note:
Full discovery is often **NP-hard**. Local MB methods are typically $O(p^2)$ or even $O(p)$ in the number of conditional independence (CI) tests.

Algorithm	Approach	Output	Strength
HITON-MB	CI-based	$\mathrm{MB}(Y)$	High-dimensional data
IAMB	CI-based	$\mathrm{MB}(Y)$	Converges fast
GES-MB	Score-based	$\mathrm{MB}(Y)$	Small sample size
linGAM	Functional	$\mathrm{Pa}(Y)$	Directs causal edges

Complexity Note:
Full discovery is often **NP-hard**. Local MB methods are typically $O(p^2)$ or even $O(p)$ in the number of conditional independence (CI) tests.

Algorithm	Approach	Output	Strength
HITON-MB	CI-based	$\mathrm{MB}(Y)$	High-dimensional data
IAMB	CI-based	$\mathrm{MB}(Y)$	Converges fast
GES-MB	Score-based	$\mathrm{MB}(Y)$	Small sample size
linGAM	Functional	$\mathrm{Pa}(Y)$	Directs causal edges

Takeaway

Use **CI-based** or **score-based** methods when only MB is needed; use **Functional** methods when you need to distinguish *causes* from *effects* and *spouses*.

Causality-based Feature Selection: Methods and Evaluations

KUI YU and XIANJIE GUO, Hefei University of Technology, China
LIN LIU and JIUYONG LI, University of South Australia, Australia
HAO WANG and ZHAOLONG LING, Hefei University of Technology, China
XINDONG WU, Mininglamp Technology, China

Feature selection is a crucial preprocessing step in data analytics and machine learning. Classical feature selection algorithms select features based on the correlations between predictive features and the class variable and do not attempt to capture causal relationships between them. It has been shown that the knowledge about the causal relationships between features and the class variable has potential benefits for building interpretable and robust prediction models, since causal relationships imply the underlying mechanism of a system. Consequently, causality-based feature selection has gradually attracted greater attentions and many algorithms have been proposed. In this article, we present a comprehensive review of recent advances in causality-based feature selection. To facilitate the development of new algorithms in the research area and make it easy for the comparisons between new methods and existing ones, we develop the first open-source package, called CausalFS, which consists of most of the representative causality-based feature selection algorithms (available at https://github.com/kuiy/CausalFS). Using CausalFS, we conduct extensive experiments to compare the representative algorithms with both synthetic and real-world datasets. Finally, we discuss some challenging problems to be tackled in future research.

CCS Concepts: • **Computing methodologies → Feature selection**;

Additional Key Words and Phrases: Feature selection, Bayesian network, Markov boundary

ACM Reference format:
Kui Yu, XianJie Guo, Lin Liu, Jiuyong Li, Hao Wang, Zhaolong Ling, and Xindong Wu. 2020. Causality-based Feature Selection: Methods and Evaluations. *ACM Comput. Surv.* 53, 5, Article 111 (September 2020), 36 pages.
https://doi.org/10.1145/3409382

111

Yu et al. (2020)

Causal Feature Selection Review

Causality-based Feature Selection: Methods and Evaluations

KUI YU and XIANJIE GUO, Hefei University of Technology, China
LIN LIU and JIUYONG LI, University of South Australia, Australia
HAO WANG and ZHAOLONG LING, Hefei University of Technology, China
XINDONG WU, Mininglamp Technology, China

Feature selection is a crucial preprocessing step in data analytics and machine learning. Classical feature selection algorithms select features based on the correlations between predictive features and the class variable and do not attempt to capture causal relationships between them. It has been shown that the knowledge about the causal relationships between features and the class variable has potential benefits for building interpretable and robust prediction models, since causal relationships imply the underlying mechanism of a system. Consequently, causality-based feature selection has gradually attracted greater attentions and many algorithms have been proposed. In this article, we present a comprehensive review of recent advances in causality-based feature selection. To facilitate the development of new algorithms in the research area and make it easy for the comparisons between new methods and existing ones, we develop the first open-source package, called CausalFS, which consists of most of the representative causality-based feature selection algorithms (available at https://github.com/kuiy/CausalFS). Using CausalFS, we conduct extensive experiments to compare the representative algorithms with both synthetic and real-world datasets. Finally, we discuss some challenging problems to be tackled in future research.

CCS Concepts: • **Computing methodologies** → **Feature selection**;

Additional Key Words and Phrases: Feature selection, Bayesian network, Markov boundary

ACM Reference format:
Kui Yu, XianJie Guo, Lin Liu, Jiuyong Li, Hao Wang, Zhaolong Ling, and Xindong Wu. 2020. Causality-based Feature Selection: Methods and Evaluations. *ACM Comput. Surv.* 53, 5, Article 111 (September 2020), 36 pages.
https://doi.org/10.1145/3409382

111

Yu et al. (2020)

Causal Feature Selection Review

- Main definitions and assumptions for causal discovery and causal feature selection

Causality-based Feature Selection: Methods and Evaluations

KUI YU and XIANJIE GUO, Hefei University of Technology, China
LIN LIU and JIUYONG LI, University of South Australia, Australia
HAO WANG and ZHAOLONG LING, Hefei University of Technology, China
XINDONG WU, Mininglamp Technology, China

Feature selection is a crucial preprocessing step in data analytics and machine learning. Classical feature selection algorithms select features based on the correlations between predictive features and the class variable and do not attempt to capture causal relationships between them. It has been shown that the knowledge about the causal relationships between features and the class variable has potential benefits for building interpretable and robust prediction models, since causal relationships imply the underlying mechanism of a system. Consequently, causality-based feature selection has gradually attracted greater attentions and many algorithms have been proposed. In this article, we present a comprehensive review of recent advances in causality-based feature selection. To facilitate the development of new algorithms in the research area and make it easy for the comparisons between new methods and existing ones, we develop the first open-source package, called CausalFS, which consists of most of the representative causality-based feature selection algorithms (available at https://github.com/kuiy/CausalFS). Using CausalFS, we conduct extensive experiments to compare the representative algorithms with both synthetic and real-world datasets. Finally, we discuss some challenging problems to be tackled in future research.

CCS Concepts: • **Computing methodologies** → **Feature selection**;

Additional Key Words and Phrases: Feature selection, Bayesian network, Markov boundary

ACM Reference format:
Kui Yu, XianJie Guo, Lin Liu, Jiuyong Li, Hao Wang, Zhaolong Ling, and Xindong Wu. 2020. Causality-based Feature Selection: Methods and Evaluations. *ACM Comput. Surv.* 53, 5, Article 111 (September 2020), 36 pages.
https://doi.org/10.1145/3409382

111

Yu et al. (2020)

Causal Feature Selection Review

- Main definitions and assumptions for causal discovery and causal feature selection
- Systematic description of the different algorithms and their general strategy

4 Outline

1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

Well defined interventions

Intervention implies full control over a variable and independence with the downstream mechanisms

Well defined interventions

Intervention implies full control over a variable and independence with the downstream mechanisms

In climate science, variables are **tightly coupled**:

Well defined interventions

Intervention implies full control over a variable and independence with the downstream mechanisms

In climate science, variables are **tightly coupled**:

- Independence of Causes and Mechanisms often fail.

Well defined interventions

Intervention implies full control over a variable and independence with the downstream mechanisms

In climate science, variables are **tightly coupled**:

- Independence of Causes and Mechanisms often fail.
- Interventions on high level emergent phenomenon are often ill-defined.

Well defined interventions

Intervention implies full control over a variable and independence with the downstream mechanisms

In climate science, variables are **tightly coupled**:

- Independence of Causes and Mechanisms often fail.
- Interventions on high level emergent phenomenon are often ill-defined.
- We cannot intervene on ENSO without simultaneously altering its relationship with atmospheric conditions.

4 The ICP Algorithm

Input: data from multiple environments $e \in \mathcal{E}$

4 The ICP Algorithm

Input: data from multiple environments $e \in \mathcal{E}$

1. **For each candidate set** $S \subseteq V \setminus \{Y\}$, test in

$$Y^e = \beta^\top X_S^e + \epsilon^e$$

if coefficients and residual distribution ϵ^e are stable across *all* environments e.

4 The ICP Algorithm

Input: data from multiple environments $e \in \mathcal{E}$

1. **For each candidate set** $S \subseteq V \setminus \{Y\}$, test in

$$Y^e = \beta^\top X_S^e + \epsilon^e$$

   if coefficients and residual distribution ϵ^e are stable across *all* environments e.

2. **Collect** all S that *pass* the invariance test.

4 The ICP Algorithm

Input: data from multiple environments $e \in \mathcal{E}$

1. **For each candidate set** $S \subseteq V \setminus \{Y\}$, test in

$$Y^e = \beta^\top X_S^e + \epsilon^e$$

   if coefficients and residual distribution ϵ^e are stable across *all* environments e.

2. **Collect** all S that *pass* the invariance test.
3. **Output:** intersection of all passing sets

$$\widehat{\mathrm{Pa}(Y)} = \bigcap \{S : \text{invariant across all } e\}$$

4 The ICP Algorithm

Input: data from multiple environments $e \in \mathcal{E}$

1. **For each candidate set** $S \subseteq V \setminus \{Y\}$, test in

$$Y^e = \beta^\top X_S^e + \epsilon^e$$

   if coefficients and residual distribution ϵ^e are stable across *all* environments e.
2. **Collect** all S that *pass* the invariance test.
3. **Output:** intersection of all passing sets

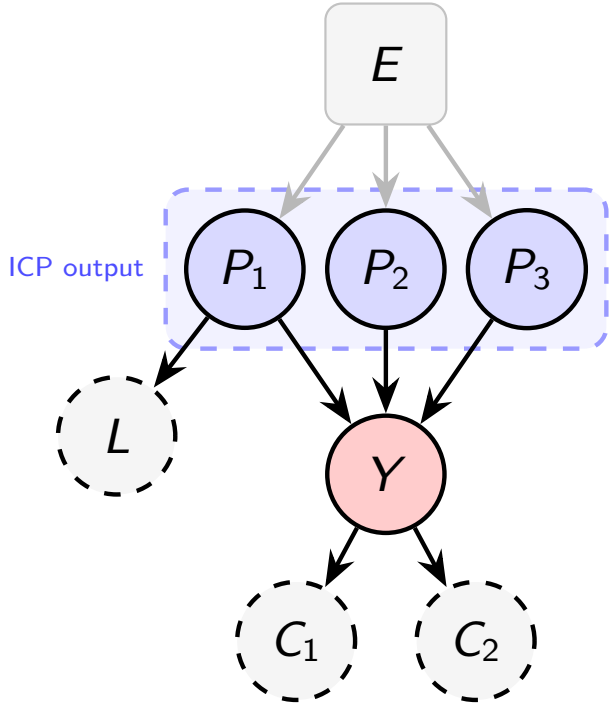
$$\widehat{\mathrm{Pa}(Y)} = \bigcap \{S : \text{invariant across all } e\}$$

Guarantee

With high probability, $\widehat{\mathrm{Pa}(Y)} \subseteq \mathrm{Pa}(Y)$ — no false positives.
ICP may miss some parents (low power), but the parents it finds are reliable.

Natural environments are already available

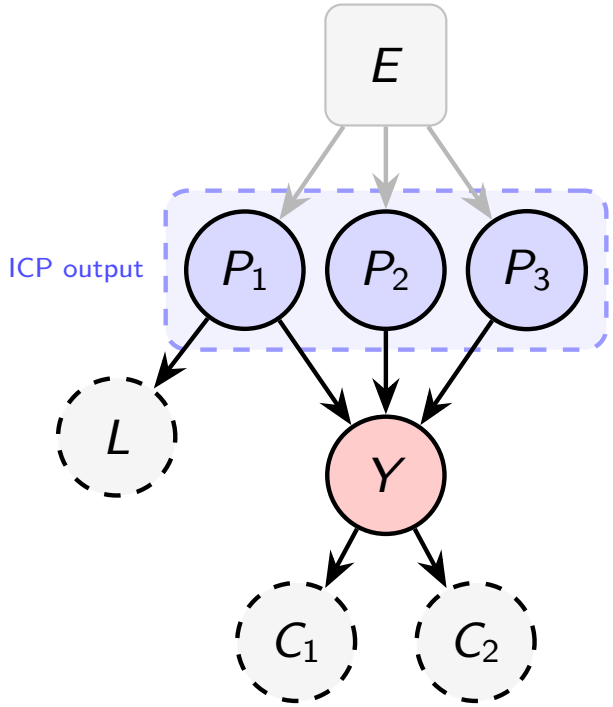
- Different **seasons** (DJF vs. JJA)



$P(Y \mid \mathrm{PA}(Y))$ *is stable across environments*

Natural environments are already available

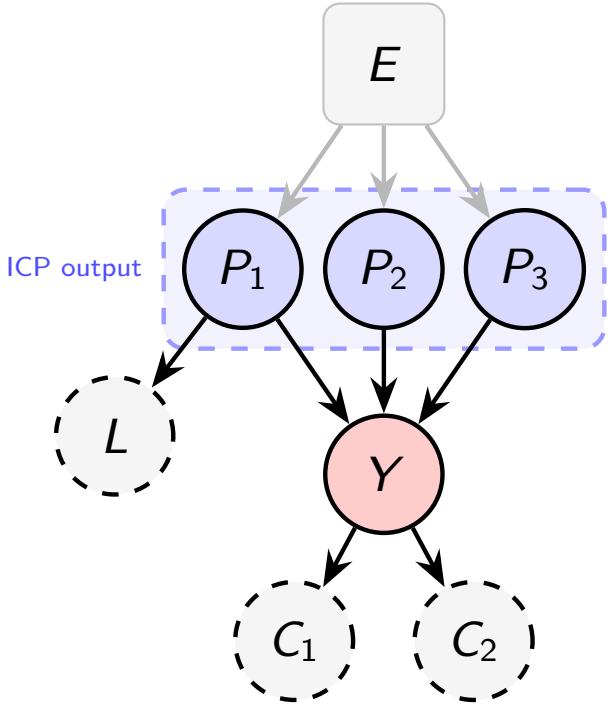
- Different **seasons** (DJF vs. JJA)
- Different **geographic regions**



$P(Y \mid \mathrm{PA}(Y))$ *is stable across environments*

Natural environments are already available

- Different **seasons** (DJF vs. JJA)
- Different **geographic regions**
- CMIP6 **scenario ensemble members** (SSP1-2.6 vs. SSP5-8.5)



$P(Y \mid \mathrm{PA}(Y))$ *is stable across environments*

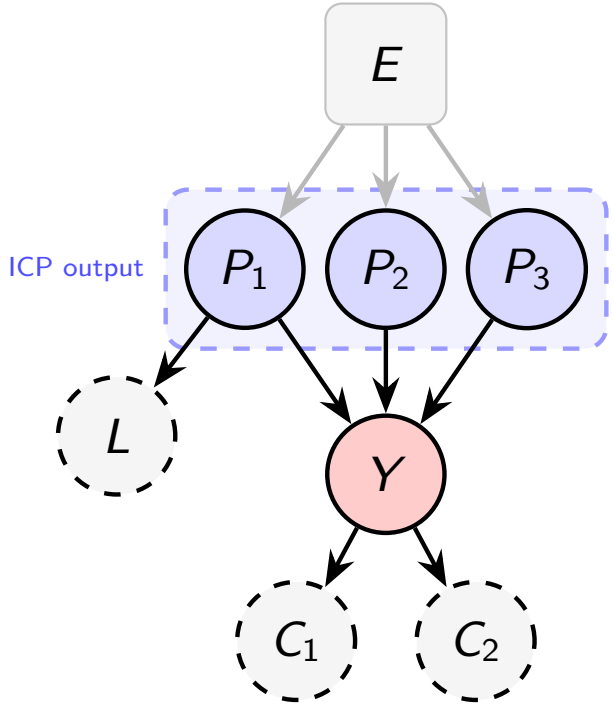
Natural environments are already available

- Different **seasons** (DJF vs. JJA)
- Different **geographic regions**
- CMIP6 **scenario ensemble members** (SSP1-2.6 vs. SSP5-8.5)

No need to define or run explicit interventions.

Directly answers the robustness question

- Output the features robust across all environments



$P(Y \mid \mathrm{PA}(Y))$ *is stable across environments*

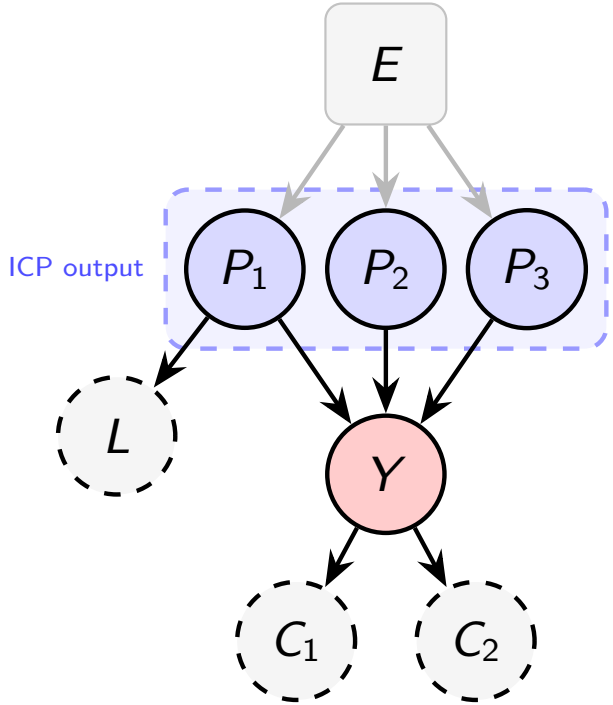
Natural environments are already available

- Different **seasons** (DJF vs. JJA)
- Different **geographic regions**
- CMIP6 **scenario ensemble members** (SSP1-2.6 vs. SSP5-8.5)

No need to define or run explicit interventions.

Directly answers the robustness question

- Output the features robust across all environments
- No need to recover the full graph first.



$P(Y \mid \mathrm{PA}(Y))$ *is stable across environments*

Beyond ICP: Can we learn *representations* that are invariant across environments, even in deep networks?

Beyond ICP: Can we learn *representations* that are invariant across environments, even in deep networks?

Invariant Risk Minimisation (IRM)

Learn a feature map Φ and a classifier w such that $w \circ \Phi$ is *simultaneously optimal* in all environments $e \in \mathcal{E}$:

$$\min_{\Phi,\, w} \sum_{e \in \mathcal{E}} \mathcal{L}^e(w \circ \Phi) \quad \text{s.t.} \quad w \in \arg\min_{\tilde{w}} \mathcal{L}^e(\tilde{w} \circ \Phi)\ \forall e$$

Beyond ICP: Can we learn *representations* that are invariant across environments, even in deep networks?

Invariant Risk Minimisation (IRM)

Learn a feature map Φ and a classifier w such that $w \circ \Phi$ is *simultaneously optimal* in all environments $e \in \mathcal{E}$:

$$\min_{\Phi,\, w} \sum_{e \in \mathcal{E}} \mathcal{L}^e(w \circ \Phi) \quad \text{s.t.} \quad w \in \arg\min_{\tilde{w}} \mathcal{L}^e(\tilde{w} \circ \Phi)\ \forall e$$

- **Causal Features:** Encourages Φ to discard spurious correlations by forcing the classifier to be optimal everywhere.

Beyond ICP: Can we learn *representations* that are invariant across environments, even in deep networks?

Invariant Risk Minimisation (IRM)

Learn a feature map Φ and a classifier w such that $w \circ \Phi$ is *simultaneously optimal* in all environments $e \in \mathcal{E}$:

$$\min_{\Phi,\, w} \sum_{e \in \mathcal{E}} \mathcal{L}^e(w \circ \Phi) \quad \text{s.t.} \quad w \in \arg\min_{\tilde{w}} \mathcal{L}^e(\tilde{w} \circ \Phi)\ \forall e$$

- **Causal Features:** Encourages Φ to discard spurious correlations by forcing the classifier to be optimal everywhere.
- Very Similar to ICP but learns a **feature representation** instead of selecting features

Anchor Regression

Instruments ("anchors") A affect X and Y. Anchor regression interpolates between OLS and Instrumental Variables (IV):

$$\hat{\beta}(\gamma) = \arg\min_{\beta} \mathbb{E}[(Y - X\beta)^2] + (\gamma - 1)\mathbb{E}[(P_A(Y - X\beta))^2]$$

where P_A projects onto the anchor space (exogenous causes).

Anchor Regression

Instruments ("anchors") A affect X and Y. Anchor regression interpolates between OLS and Instrumental Variables (IV):

$$\hat{\beta}(\gamma) = \arg\min_{\beta} \mathbb{E}[(Y - X\beta)^2] + (\gamma - 1)\mathbb{E}[(P_A(Y - X\beta))^2]$$

where P_A projects onto the anchor space (exogenous causes).

- $\gamma = 0$: Ordinary OLS (best in-sample).

Anchor Regression

Instruments ("anchors") A affect X and Y. Anchor regression interpolates between OLS and Instrumental Variables (IV):

$$\hat{\beta}(\gamma) = \arg\min_{\beta} \mathbb{E}[(Y - X\beta)^2] + (\gamma - 1)\mathbb{E}[(P_A(Y - X\beta))^2]$$

where P_A projects onto the anchor space (exogenous causes).

- $\gamma = 0$: Ordinary OLS (best in-sample).
- $\gamma \to \infty$: Two-stage IV (maximally robust).

Anchor Regression

Instruments ("anchors") A affect X and Y. Anchor regression interpolates between OLS and Instrumental Variables (IV):

$$\hat{\beta}(\gamma) = \arg\min_{\beta} \mathbb{E}[(Y - X\beta)^2] + (\gamma - 1)\mathbb{E}[(P_A(Y - X\beta))^2]$$

where P_A projects onto the anchor space (exogenous causes).

- $\gamma = 0$: Ordinary OLS (best in-sample).
- $\gamma \to \infty$: Two-stage IV (maximally robust).
- **Tuning γ:** *accuracy–robustness trade-off.*

Anchor Regression

Instruments ("anchors") A affect X and Y. Anchor regression interpolates between OLS and Instrumental Variables (IV):

$$\hat{\beta}(\gamma) = \arg\min_{\beta} \mathbb{E}[(Y - X\beta)^2] + (\gamma - 1)\mathbb{E}[(P_A(Y - X\beta))^2]$$

where P_A projects onto the anchor space (exogenous causes).

- $\gamma = 0$: Ordinary OLS (best in-sample).
- $\gamma \to \infty$: Two-stage IV (maximally robust).
- **Tuning γ:** *accuracy–robustness trade-off.*

The Core Insight

Enforcing invariance across environments implicitly selects causal features and discards spurious correlations — without requiring explicit causal graph recovery.

5 Outline

1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

5 Take-Aways

1. **Standard ERM** learns every correlation — causal and spurious. Under distribution shift (scenarios, tipping points, new regions) it can fail.

5 Take-Aways

1. **Standard ERM** learns every correlation — causal and spurious. Under distribution shift (scenarios, tipping points, new regions) it can fail.
2. **Robust ML** (minimax ERM over $\mathcal{P}$) finds predictors that are safe across a range of plausible distributions.

5 Take-Aways

1. **Standard ERM** learns every correlation — causal and spurious. Under distribution shift (scenarios, tipping points, new regions) it can fail.
2. **Robust ML** (minimax ERM over $\mathcal{P}$) finds predictors that are safe across a range of plausible distributions.
3. **Causal structure determines the optimal robust predictor**: parents $\mathrm{Pa}(Y)$ for maximal robustness; Markov Blanket $\mathrm{MB}(Y)$ for accuracy when shifts are moderate.

5 Take-Aways

1. **Standard ERM** learns every correlation — causal and spurious. Under distribution shift (scenarios, tipping points, new regions) it can fail.
2. **Robust ML** (minimax ERM over $\mathcal{P}$) finds predictors that are safe across a range of plausible distributions.
3. **Causal structure determines the optimal robust predictor**: parents $\mathrm{Pa}(Y)$ for maximal robustness; Markov Blanket $\mathrm{MB}(Y)$ for accuracy when shifts are moderate.
4. **Causal discovery and Feature Selection** can recover the graph (or equivalence class) from data — but some arrows might remain ambiguous without extra assumptions.

5 Take-Aways

1. **Standard ERM** learns every correlation — causal and spurious. Under distribution shift (scenarios, tipping points, new regions) it can fail.
2. **Robust ML** (minimax ERM over $\mathcal{P}$) finds predictors that are safe across a range of plausible distributions.
3. **Causal structure determines the optimal robust predictor**: parents $\mathrm{Pa}(Y)$ for maximal robustness; Markov Blanket $\mathrm{MB}(Y)$ for accuracy when shifts are moderate.
4. **Causal discovery and Feature Selection** can recover the graph (or equivalence class) from data — but some arrows might remain ambiguous without extra assumptions.
5. **ICP** leverages multiple environments (natural in climate science) to identify $\mathrm{Pa}(Y)$ directly, without recovering the full graph.

5 Take-Aways

1. **Standard ERM** learns every correlation — causal and spurious. Under distribution shift (scenarios, tipping points, new regions) it can fail.
2. **Robust ML** (minimax ERM over $\mathcal{P}$) finds predictors that are safe across a range of plausible distributions.
3. **Causal structure determines the optimal robust predictor**: parents $\mathrm{Pa}(Y)$ for maximal robustness; Markov Blanket $\mathrm{MB}(Y)$ for accuracy when shifts are moderate.
4. **Causal discovery and Feature Selection** can recover the graph (or equivalence class) from data — but some arrows might remain ambiguous without extra assumptions.
5. **ICP** leverages multiple environments (natural in climate science) to identify $\mathrm{Pa}(Y)$ directly, without recovering the full graph.
6. **Domain knowledge is indispensable**: methods structure and formalise it, but cannot replace it.

5 Take-Aways

1. **Standard ERM** learns every correlation — causal and spurious. Under distribution shift (scenarios, tipping points, new regions) it can fail.
2. **Robust ML** (minimax ERM over $\mathcal{P}$) finds predictors that are safe across a range of plausible distributions.
3. **Causal structure determines the optimal robust predictor**: parents $\mathrm{Pa}(Y)$ for maximal robustness; Markov Blanket $\mathrm{MB}(Y)$ for accuracy when shifts are moderate.
4. **Causal discovery and Feature Selection** can recover the graph (or equivalence class) from data — but some arrows might remain ambiguous without extra assumptions.
5. **ICP** leverages multiple environments (natural in climate science) to identify $\mathrm{Pa}(Y)$ directly, without recovering the full graph.
6. **Domain knowledge is indispensable**: methods structure and formalise it, but cannot replace it.

Software: `causal-learn` (PC, GES, ANM) · `tigramite` (PCMCI) · `InvariantCausalPrediction` (ICP, R)